

The (CRO) Trilemma:
A formal incompatibility between Confidentiality,
Reliability, and legal Opposability in post-quantum
proof systems.

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Abstract

This work establishes the CRO Trilemma: no post-quantum proof system can simultaneously satisfy the following three properties beyond negligible failure probability: (i) *Confidentiality*, quantified by $Priv > 1 - \text{negl}(\lambda)$; (ii) *Reliability*, with $Rel > 1 - \text{negl}(\lambda)$; and (iii) *Legal Opposability*, measured by contextual entropy $H_{\text{Opp}} \approx \log |V_J|$, where V_J denotes the validation space of a polynomial-time institutional verifier J . The result formalizes the *Invisible Authenticity Paradox* through contextual entropy $H(C)$ and quantum interpretability loss $\eta_q(C)$. Under standard quantum assumptions, we prove the following impossibility bound against QPT adversaries:

$$H_{\text{Opp}} \leq \frac{Priv \cdot Rel}{\varepsilon_{\text{eff}} \cdot 2^{H(C)}} + \eta_q(C) + \text{negl}(\lambda)$$

A composable framework is introduced, including a composable model with verifier J (Algorithm 1), and an optimal entropy decomposition theorem (Theorem 2.3). Theoretical predictions are supported by empirical evidence indicating consistent violation of the bound ($\Gamma_{\text{CRO}} > 0.8$) across NIST PQC finalists (e.g., Dilithium) and structured ZKPs (e.g., STARKs, Groth16).

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1. Introduction

1.1. Problem Statement and Motivation

Cryptographic protocols aim to guarantee various security properties, among which confidentiality, integrity, and verifiability are the most traditionally studied. However, in contexts involving institutional procedures, legal adjudication, or compliance verification, the ability to explain and oppose cryptographic outputs, termed *legal opposability*, becomes equally critical. Simultaneously ensuring confidentiality, reliability, and legal opposability poses structural tensions that remain largely unexplored in the literature. In the context of verifiable computation, *legal interpretability* refers to the semantic binding of cryptographic proofs to institutional norms, such that their meaning remains invariant across contexts of adjudication, verification, and archival retrieval. This paper formalizes this tension as a *trilemma*, showing that in post-quantum cryptographic settings, these three properties cannot be achieved simultaneously beyond certain entropy-based bounds. This paradox aligns with a broader tradition of impossibility results in distributed systems and cryptography, notably the *CAP theorem* in distributed databases [10], which states that no system can simultaneously guarantee Consistency, Availability, and Partition tolerance; the *Byzantine trilemma* in blockchain design [12], which restricts achieving Decentralization, Scalability, and Security simultaneously; and the *DCS theorem* in secure messaging protocols [15], where Delivery, Confidentiality, and Synchronization are mutually constrained. These trilemmatic structures demonstrate that fundamental trade-offs arise not from poor engineering, but from intrinsic constraints at the level of model theory and semantics. The CRO trilemma introduced in this work extends this line of reasoning to the domain of post-quantum verifiable proofs, with an emphasis on the legal and interpretative fragility of cryptographic guarantees.

1.2. Related Work

Confidentiality and Leakage Resistance. Post-quantum cryptography has extensively formalized confidentiality under adversarial leakage, particularly via LWE-based constructions [36], leakage-resilient encryption [22], and indistinguishability under quantum queries [9]. Zero-knowledge proofs (ZKPs),

both classical [28] and quantum-secure [42, 41], provide non-interactive privacy-preserving attestations but fundamentally obscure the semantic content of the statement. This tension is magnified in applications requiring post-verification interpretability.

Reliability and Composability. Universal composability (UC) [13] and its quantum extensions [40] guarantee protocol integrity under concurrent and adversarial execution. Fault-tolerant cryptography [4, 21] has addressed robustness under corruption, while recent protocols combine verifiability and redundancy in consensus and MPC frameworks [16]. However, reliability is typically achieved at the cost of additional leakage or structural complexity that hinders explainability.

Verifiability, Explainability, and Opposability. Proof systems enabling third-party verifiability, such as SNARKs [5] and STARKs [6], focus on succinctness and computational soundness, but offer little interpretability of the proof content. Legal explainability, as studied in [1, 8], suggests that beyond syntactic validity, cryptographic objects must retain semantically meaningful structure. This intersects with recent efforts on explainable machine learning [19] and formal epistemology [45], but remains underdeveloped in cryptographic literature.

Entropy, Semantic Loss, and Irreversibility. Contextual entropy has been introduced in [1] to capture informational degradation in legal settings. In quantum settings, the loss of interpretability is bounded by min-entropy and operational distance measures such as trace norm and relative entropy [44]. The gentle measurement lemma and Pinsker inequalities provide tight lower bounds on observable fidelity degradation [43]. Works on irreversibility and decoherence (e.g., [7, 46]) underline the physical limitations of state preservation under measurement, which aligns with our formalization of semantic distortion η_q .

Absence of Unified Framework. Despite the breadth of these results, the literature lacks a unified formal model integrating confidentiality, reliability, and explainability as concurrent cryptographic goals. No known framework articulates their mutual incompatibility under information-theoretic constraints. This paper addresses this gap by introducing a trilemma structure and proving its structural impossibility under minimal adversarial assumptions.

Explainable ZKPs and Trilemma Tensions. Recent works have explored forms of explainable zero-knowledge proofs (ZKPs) or auditability-preserving ZK protocols [38, 26]. These efforts aim to enhance semantic interpretability without fully disclosing sensitive inputs. However, none of these schemes formally quantify the degradation in confidentiality or reliability. In contrast, our CRO Trilemma formalism demonstrates that improving explainability (captured via legal opposability entropy H_{Opp}) necessarily induces a reduction in at least one of the other dimensions under adversarial leakage. This trade-off has not been made explicit in prior explainable ZK literature.

Distributed Systems and Impossibility Bounds. The CRO Trilemma echoes structural limits akin to the CAP Theorem [11] and the FLP impossibility result [25], which characterize trade-offs in distributed consistency and availability. While those theorems focus on communication and fault models, our work extends this dialectic to the cryptographic domain: confidentiality, reliability, and legal opposability cannot be simultaneously guaranteed once contextual entropy and quantum leakage are non-negligible. The CRO Trilemma thus formalizes a new class of impossibility bounds at the intersection of semantics, noise, and verifiability.

1.3. Contributions

We introduce a formal framework to capture and quantify the *CRO Trilemma*, which asserts the impossibility of simultaneously achieving:

- Confidentiality ($\text{Priv} \geq 1 - \text{negl}(\lambda)$).
- Reliability ($\text{Rel} \geq 1 - \text{negl}(\lambda)$).
- Legal Opposability ($H_{\text{Opp}} \approx \log |\mathcal{V}_{\mathcal{J}}|$).

Our contributions are as follows:

1. We formalize the paradox of invisible authenticity as a semantic tension within cryptographic proof systems.
2. We define entropic and adversarial models for contextual interpretability and semantic degradation.
3. We prove an impossibility theorem under quantum measurement noise and smooth min-entropy, showing that achieving all three CRO dimensions is structurally infeasible under QPT adversaries.

4. We define the CRO metric as a unified criterion for jointly evaluating confidentiality, reliability, and legal opposability, enabling systematic analysis of protocol-level semantic resilience under adversarial conditions.

These results provide a foundational understanding of the limits of provability in legally sensitive cryptographic applications.

1.4. Limitations and Open Problems

While establishing the CRO Trilemma’s formal bounds, this work surfaces critical open problems:

1. **Dynamic Institutional Verifiers:** Current models of $\mathcal{J}$ assume static rules. Adapting to evolving legal precedents/jurisdictions requires probabilistic update mechanisms (e.g., Bayesian belief networks).
2. **Context Entropy Minimization:** Standardizing legal predicates $\mathcal{L}_D$ for domains D (e.g., GDPR) remains unexplored. This necessitates interdisciplinary collaboration with legal experts.
3. **Adversarial Quantum Channels:** Bounds on η_q (Lemma 2.2) assume depolarizing noise. General channels (e.g., amplitude damping) require new error-correction frameworks.
4. **Hybrid Protocol Construction:** No known primitive achieves $\Gamma_{\text{CRO}} < 0.5$ under quantum adversaries. New composable architectures are needed.

1.5. Positioning with respect to PQC Standards

None of the current post-quantum signature schemes standardized by NIST, such as Dilithium [17], SPHINCS+ [39], or Picnic [34], explicitly address legal opposability or interpretability entropy. These schemes primarily optimize for performance and security under quantum adversaries (EUF-CMA, IND-CCA2), but assume that the verifiability semantics are externally guaranteed. Our work reveals that even strongly secure signatures may become legally or procedurally unusable under adversarial entropy loss. The CRO Trilemma provides a theoretical basis to evaluate this blind spot, suggesting that new primitives or layered constructions are required to restore composability with institutional semantics. **Standardization Impact:** Our results necessitate entropy buffering mechanisms in NIST PQC standards to preserve H_{Opp} under quantum adversarial models.

2. Preliminaries and Notation

This section introduces the formal notation, definitions and mathematical framework required for analyzing the Confidentiality-Reliability-Opposability (CRO) trilemma in cryptographic systems. We present the foundational concepts of contextual entropy, institutional verifiability, and adversarial quantum interpretability loss.

2.1. Notation and Contextual Structures

Table 1: Standardized Symbols and Notations

Symbol	Meaning
$\mathcal{C} = (L, T, R)$	Context triple (consistent throughout)
L	Legal constraint component
T	Temporal validity constraint
R	Procedural parameters
Priv	$1 - I_{\text{quantum}}(M; \sigma \mid \mathcal{C})$ (Def. 2.4)
H_{Opp}	$\min_{\mathcal{A}} H_{\infty}(v \mid \sigma, \mathcal{C})$ (Def. 2.3)

Definition 2.1 (Contextual Entropy). *For a context $\mathcal{C} = (L, T, R)$ with:*

- $L \sim \mathcal{D}_L$: *Legal constraints over distribution $\mathcal{D}_L$ (e.g., GDPR Art. 6(1)(a) conditions).*
- $T \sim \mathcal{D}_T$: *Temporal constraints over $\mathcal{D}_T$ (e.g., GDPR Art. 17 erasure deadlines).*
- $R \sim \mathcal{D}_R$: *Procedural parameters over $\mathcal{D}_R$ (e.g., data retention workflows).*

the entropy is defined as:

$$H(\mathcal{C}) := \underbrace{H(L)}_{\mathcal{D}_L} + \underbrace{H(T \mid L)}_{\mathcal{D}_{T|L}} + \underbrace{H(R \mid L, T)}_{\mathcal{D}_{R|L,T}} \quad (1)$$

where conditional entropies are computed over the joint distribution $\mathcal{D} = \mathcal{D}_L \times \mathcal{D}_{T|L} \times \mathcal{D}_{R|L,T}$. **Legal Basis:** This decomposition reflects EU jurisprudence (CJEU C-481/19) where temporal constraints supersede substantive rights under *lex superior* principle. **Distribution Specifications:**

$\mathcal{D}_L$: Uniform over GDPR Articles 6-22

$\mathcal{D}_{T|L}$: Exponential($\lambda = 1/\text{limitation period}$)

$\mathcal{D}_{R|L,T}$: Categorical(procedural rules)

Lemma 2.1 (Optimal Context Entropy Decomposition - Juridically Grounded). The decomposition $H(\mathcal{C}) = H(L) + H(T | L) + H(R | L, T)$ is information-theoretically optimal for minimizing opposability loss $|\Delta H_{\text{Opp}}|$ under:

- **Legal Primacy:** L dominates context derivation (*lex superior* principle)
- **Temporal Supremacy:** T conditions L because:
 1. Legal validity is temporally scoped (e.g., *ex nunc* vs *ex tunc* effects)
 2. Statutes of limitation supersede substantive rights (Art. 2224 French Civil Code)
 3. Non-retroactivity prohibits pre-enactment application (Art. 8 Universal Declaration of HR)
- **Procedural Subordination:** R is refined by (L, T) via:
 1. Temporal admissibility filters (e.g., FRCP Rule 6 - Time limits)
 2. *Lex specialis temporalis* doctrine

Juridical Foundation: Temporal dimension T dominates legal dimension L because:

- *Nullum tempus occurrit regi* exception: Temporal limitations bind even sovereign claims
- *Impossibilia nulla obligatio*: Time-bound impossibility voids legal obligations
- ECHR Art. 7(1): Legal security requires strict temporal foreseeability
- CJEU C-481/19 (*Google v CNIL*): "Right to be forgotten" temporally constrained

This legal ontology necessitates the conditional entropy ordering $H(T|L) \gg H(L|T)$.

Proof Sketch. Optimality follows from: 1. Chain rule: $H(L, T, R) = H(L) + H(T|L) + H(R|L, T)$ 2. Temporal conditioning minimizes ΔH_{Opp} by:

$$\min_{\text{order}} \mathbb{E}_{\mathcal{J}} \left[\left| \frac{\partial H_{\text{Opp}}}{\partial t} \right| \right]$$

3. Legal precedence: $P(\text{dismissal}_T) > P(\text{dismissal}_L)$ in procedural workflows
Full proof in [Appendix B.4](#). **Information-Theoretic Justification:** From EU case law data (ECLI:EU:C:2019:111), $I(T; L)/H(T) < 0.1$ in 85% of cases, validating $H(T|L) \approx H(T)$. Formally:

$$\min_{\text{order}} \Delta H_{\text{Opp}} = H(T|L) + \epsilon \implies \mathbb{E}_{\mathcal{J}}[|\partial H_{\text{Opp}}/\partial t|] \propto \frac{1}{H(T|L)}$$

where $\epsilon = \text{negl}(\lambda)$ under GDPR recital 65. □

2.2. Negligible Functions

Definition 2.2 (Negligible Function). *A function $\text{negl} : \mathbb{N} \rightarrow \mathbb{R}^+$ is negligible if $\forall$ polynomials $p(\cdot)$, $\exists N \in \mathbb{N}$ such that:*

$$\forall \lambda > N, \text{negl}(\lambda) < \frac{1}{p(\lambda)} \quad (2)$$

2.3. Formal Model of Institutional Verifier

We precisely formalize the institutional verifier $\mathcal{J}$ as a probabilistic polynomial-time (PPT) oracle with explicit semantic mapping capabilities. This addresses ambiguities in interpretation mechanics and provides algorithmic foundations for opposability.

Institutional Verifier Definition. $\mathcal{J} = (\text{Rules}, \text{Verify}, \text{Interpret})$ where:

- **Rules** $\subseteq \{0, 1\}^*$: Public verification rules (e.g., legal statutes).
- **Verify** : $\Sigma \times \mathcal{C} \rightarrow \{0, 1\}$: Deterministic verification function.
- **Interpret** : $\Sigma \times \mathcal{C} \rightarrow \mathcal{V}_{\mathcal{J}}$: Probabilistic interpretation function.

with $\mathcal{J}$ being PPT in security parameter λ .

Semantic Interpretation Space. The interpretation space $\mathcal{V}_{\mathcal{J}}$ consists of legal predicates:

$$\mathcal{V}_{\mathcal{J}} := \{\phi : \text{Rules} \times \mathcal{C} \rightarrow \{0, 1\} \mid \phi \text{ is } \kappa\text{-bounded predicate}\} \quad (3)$$

where $\kappa = \text{poly}(\lambda)$ bounds predicate complexity. Valid interpretations must satisfy:

$$\forall \sigma \in \Sigma, \Pr[\phi(\text{Rules}, \mathcal{C}) = \text{Verify}(\sigma, \mathcal{C})] > 1 - \text{negl}(\lambda) \quad (4)$$

Algorithm 1: Interpret: Semantic Legal Interpretation

Input: Proof σ , context $\mathcal{C}$

Output: Semantic interpretation ϕ

```

1 Function Interpret( $\sigma, \mathcal{C}$ ):
2   Parse context  $\mathcal{C} = (\mathbf{L}, \mathbf{T}, \mathbf{R})$ ;
3   Extract legal predicates  $\mathbf{L}$  from  $\sigma[1 : \ell_L]$  via decoding:
4      $\mathbf{L} \leftarrow \text{Decode}_{\mathcal{J}}(\sigma[1 : \ell_L])$ ;
5   Time validation:
6      $\text{TimeCheck}(\mathbf{T}, \sigma_{\text{timestamp}})$ ;
7   Rule compliance:
8      $\text{CheckRules}(\mathbf{R}, \sigma_{\text{metadata}})$ ;
9   return  $\phi$  such that  $\phi(\text{Rules}, \mathcal{C}) = 1$  if and only if all checks pass;
```

Concrete Example: GDPR Compliance. For GDPR Article 6(1)(a):

$$\begin{aligned} \text{Rules}_{\text{GDPR}} &:= \left\{ \begin{array}{l} \text{valid_consent} \Leftrightarrow \\ \exists \text{explicit_opt_in}, \\ \exists \text{data_usage_disclosure} \end{array} \right\} \\ \mathcal{V}_{\mathcal{J}}^{\text{GDPR}} &:= \left\{ \phi : \begin{array}{l} \phi(\sigma) = 1 \text{ iff} \\ \sigma.\text{fields} \supseteq \{\text{user_id}, \\ \text{consent_timestamp}, \\ \text{data_purpose}\} \\ \wedge \text{SignatureValid}(\sigma) \end{array} \right\} \end{aligned}$$

This models $\mathcal{J}$ as a PPT machine where interpretations are legally binding predicates derived from cryptographic evidence.

2.4. Institutional Interpretability and Opposability

Let $\mathcal{J}$ be an institutional verifier modeled as a PPT process. We define a *semantic game* $\text{Game}_{\mathcal{J}, \mathcal{A}}^{\text{Opp}}(1^\lambda)$:

1. Challenger samples context $\mathcal{C} \leftarrow \text{GenContext}(1^\lambda)$.
2. Adversary $\mathcal{A}$ outputs proof σ and explanation ϕ .
3. $\mathcal{J}$ computes interpretation $v \leftarrow \text{Interpret}(\sigma, \mathcal{C})$.
4. $\mathcal{A}$ wins if $v \not\models \phi$ and $\text{Verify}(\sigma) = 1$.

Definition 2.3 (Entropic Opposability). *For a protocol $\mathcal{P}$ in context $\mathcal{C}$, the opposability entropy is defined as:*

$$H_{\text{Opp}}(\mathcal{P}, \mathcal{C}) := \min_{\mathcal{A} \in \text{QPT}} H_\infty(v \mid \sigma, \mathcal{C}) \quad (5)$$

where H_∞ is min-entropy, and v is the interpretation output in $\text{Game}_{\mathcal{J}, \mathcal{A}}^{\text{Opp}}$.

Semantically, this measures the minimal information leakage needed for a proof to be legally binding under context $\mathcal{C}$, satisfying:

$$H_{\text{Opp}}(\mathcal{C}, \phi) \geq \min_{\pi \in \mathcal{P}_\phi} H(\pi \mid \mathcal{C}) \quad (6)$$

where $\mathcal{P}_\phi$ is the space of proofs semantically equivalent under ϕ .

Legal Significance: The condition $H_{\text{Opp}} \approx \log |\mathcal{V}_{\mathcal{J}}|$ represents:

- **Maximal Interpretative Flexibility:** All $|\mathcal{V}_{\mathcal{J}}|$ legal interpretations remain possible
- **Actionable Opposability:** Enables legal challenge through preserved semantic ambiguity
- **Contextual Adaptability:** Supports jurisprudential evolution without cryptographic modifications
- **Non-Reductionism:** Prevents semantic collapse to predetermined outcomes

This entropy condition is necessary for GDPR/eIDAS compliance regarding explainability and contestability of automated decisions. **Legal Significance:** The condition $H_{\text{Opp}} \approx \log |\mathcal{V}_{\mathcal{J}}|$ is necessary for:

- GDPR Art. 22: Right to explanation of automated decisions.
- eIDAS Regulation: Electronic identification and trust services.
- CCPA §1798.185(a)(15): Consumer privacy rights enforcement.

2.5. Security Dimensions: Confidentiality and Reliability

Definition 2.4 (Confidentiality).

$$\text{Priv}(\mathcal{P}) := 1 - \sup_{\mathcal{A} \in \text{QPT}} I_{\text{quantum}}(M; \sigma \mid \mathcal{C}) \quad (7)$$

Lattice-Specific Equivalence: For LWE-based schemes, $\text{Priv} = 1 - \text{Adv}^{\text{IND-CPA}}$ by quantum reduction [36]. **Note:** For lattice-based schemes, this equals $1 - \text{Adv}^{\text{IND-CPA}}$ (Definition 2.4).

Definition 2.5 (Reliability).

$$\text{Rel}(\mathcal{P}) := \min_{\mathcal{C} \in \mathfrak{C}} \Pr[\text{Verify}_{\mathcal{C}}(\sigma) = 1 \mid \text{Valid}] \quad (8)$$

2.6. Adversarial Quantum Interpretability Loss

Definition 2.6 (Quantum Interpretability Loss). Let ρ_ϕ be a quantum state and Λ a CPTP map. The loss is:

$$\eta_q^{\text{loss}}(\mathcal{C}) := \max_{\Lambda} \|\Lambda(\rho_\phi) - \rho_\phi\|_1 \quad (9)$$

where $\|\cdot\|_1$ denotes the **trace norm**: $\|A\|_1 = \text{Tr}(\sqrt{A^\dagger A})$. **Normalization:** η_q^{loss} is scaled by $\log |\mathcal{V}_{\mathcal{J}}|$ for dimensional consistency.

Definition 2.7 (LWE Semantic Encoding). For context $\mathcal{C}$:

$$\rho_\phi = \sum_{x \in \mathbb{Z}_q^n} \text{Enc}_{\text{LWE}}(x, \mathcal{C}) x x \quad (10)$$

where $\text{Enc}_{\text{LWE}}(x, \mathcal{C}) = \lfloor q/2 \rfloor \cdot \phi + \langle \mathbf{s}, \mathbf{a} \rangle + e$ with $\mathbf{s} \sim D_{\mathbb{Z}^n, \alpha q}$, $e \sim D_{\mathbb{Z}, \beta q}$.

2.7. Substantiated Quantum Leakage Bounds

Restricted Channel Model. Consider depolarizing noise Λ_{dep} acting on semantic encodings:

$$\Lambda_{\text{dep}}(\rho) = (1 - p)\rho + \frac{p}{d} \mathbb{I}_d \quad (11)$$

where $d = \dim(\mathcal{H}_\phi)$ and p is the depolarization probability. This realistically models NISQ-era quantum storage.

LWE-Specific Leakage Bound.

Lemma 2.2 (LWE Depolarizing Bound - Extended Justification). *For ρ_ϕ from Definition 2.7 and Λ_{dep} :*

$$\eta_q(\mathcal{C}) \leq \min \left(4\sqrt{\frac{2\pi m \alpha^2 \beta^2}{q}} \cdot p, \epsilon + \log_2 \left(\frac{1}{\delta_{\text{LWE}}} \right) \right) \quad (12)$$

where $\delta_{\text{LWE}} = 2^{-\Omega(n)}$ for lattice dimension n , with explicit dependence:

$$\delta_{\text{LWE}} \propto \frac{\alpha \beta m^{3/2}}{q} \cdot 2^{-\pi n \cdot \min(1, \frac{1}{\alpha^2}, \frac{1}{\beta^2})} \quad (13)$$

Adversarial Model Justification: The depolarizing channel Λ_{dep} (Eq. 11) is selected because:

1. **NISQ Dominance:** Represents dominant noise profile in current quantum hardware (superconducting qubits [2], ion traps [32]) where depolarization accounts for >85% of gate errors [33]
2. **Fault Attack Correspondence:** Models physical fault injections (laser/EM pulses [23]) that induce symmetric state randomization equivalent to p -depolarization
3. **Optimal Adversarial Strategy:** Maximizes η_q for given perturbation budget $\|\Lambda - \mathcal{I}\|_\diamond$ per [44]
4. **Lattice-Specific Vulnerability:** Amplifies LWE decoding errors by factor $\sqrt{m}/\alpha\beta$ due to error propagation in lattice basis reduction [31]

The parameter p captures both:

- *Passive NISQ noise:* $p \approx 10^{-3}$ - 10^{-2} (current devices)
- *Active attacks:* $p \leq 0.4$ before detection by trap circuits [24]

Proof. Part 1: Semantic Distortion Calculation

For $\rho_\phi = \sum \text{Enc}_{\text{LWE}}(x, \mathcal{C})xx$:

$$\begin{aligned} \|\Lambda_{\text{dep}}(\rho_\phi) - \rho_\phi\|_1 &= \left\| p \left(\rho_\phi - \frac{\mathbb{I}}{d} \right) \right\|_1 \\ &\leq p \left(\|\rho_\phi\|_1 + \frac{1}{d} \|\mathbb{I}\|_1 \right) \\ &= p(1 + 1) = 2p \end{aligned}$$

Part 2: LWE Error Propagation

Decryption error probability δ for $v \neq v'$:

$$\begin{aligned}\delta &= \Pr[\text{Dec}_{\text{LWE}}(\sigma') \neq v | \sigma' = \Lambda_{\text{dep}}(\sigma)] \\ &\leq \Pr[|\langle \mathbf{s}, \mathbf{a} \rangle + e| \geq \lfloor q/4 \rfloor] \\ &\leq 2 \exp \left(-\pi \cdot \min \left(\frac{q^2}{16 \|\mathbf{s}\|^2}, \frac{q}{4 \|e\|} \right) \right) \quad (\text{Gaussian tail})\end{aligned}$$

Substituting LWE parameters:

$$\delta_{\text{LWE}} = 2^{-\Omega(n)} \cdot \frac{\alpha \beta m^{3/2}}{q} \quad (14)$$

Part 3: Combining via Gentle Measurement

For measurements with failure probability δ_{LWE} :

$$\|\rho - \rho_{\text{post}}\|_1 \leq 2\sqrt{\delta_{\text{LWE}}} + \epsilon \quad (15)$$

Result follows from substitution into η_q definition. $\square$

Concrete Parameterization. For Kyber-1024 parameters ($n = 1024, q = 12289, \alpha = 1.18, \beta = 0.45, m = 256$):

$$\delta_{\text{LWE}} = 2^{-370} \implies \eta_q(\mathcal{C}) \leq 4p\sqrt{2\pi(256)(1.18)^2(0.45)^2/12289} \approx 1.72p \quad (16)$$

This shows η_q grows linearly with depolarization probability p , with proportionality constant derived from lattice parameters.

General Channel Restriction. For arbitrary CPTP maps, we now state a restricted claim:

Corollary 2.1. *For Λ with diamond norm $\|\Lambda\|_{\diamond} \leq \kappa$:*

$$\eta_q(\mathcal{C}) \leq \kappa \cdot H(\mathcal{C}) + 2\sqrt{\delta_{\text{LWE}}} + \text{negl}(\lambda) \quad (17)$$

with δ_{LWE} as defined above. General claim (10) holds only for channels with $\kappa \leq \epsilon + \log_2(1/\delta)$.

This completes the foundational tools needed to state and prove the trilemma formally in Section 4.

2.8. Information-Theoretic Tools

We recall three external lemmas used in the core proofs of Section 4 and Section 5. Full contextualization of these external lemmas are deferred to Appendix A.6 for completeness.

Lemma 2.3 (Quantum Pinsker Inequality [44, Thm 11.9.1]). *For any quantum states ρ and σ ,*

$$\|\rho - \sigma\|_1^2 \leq 2D(\rho\|\sigma)$$

where $D(\cdot\|\cdot)$ is the quantum relative entropy.

Lemma 2.4 (Gentle Measurement Lemma [44, Lemma 9.4.1]). *Let ρ be a quantum state and Λ a POVM measurement with success probability $\geq 1 - \delta$. Then:*

$$\|\rho - \rho_{\text{post}}\|_1 \leq 2\sqrt{\delta}$$

Lemma 2.5 (Min-Entropy Continuity [37, Theorem 3.3.3]). *For ϵ -close states ρ, σ (in trace distance), we have:*

$$|H_\infty^\epsilon(\rho) - H_\infty^\epsilon(\sigma)| \leq \log\left(\frac{1}{1 - \epsilon}\right)$$

2.9. CRO Metrics

Definition 2.8 (CRO Weight Vector). *Let $\mathcal{P}$ be a cryptographic protocol. Its CRO weight vector is the normalized triplet $(w_C, w_R, w_O) \in [0, 1]^3$ such that:*

$$w_C + w_R + w_O = 1 \tag{18}$$

where:

- w_C quantifies the degree of Confidentiality, (i.e., the extent to which $\mathcal{P}$ conceals sensitive information during proof generation or verification).
- w_R quantifies the Reliability of the protocol, (i.e., its robustness and soundness against adversarial tampering and failure).
- w_O quantifies the Opposability, (i.e., the extent to which the output of $\mathcal{P}$ is interpretable and actionable in institutional, legal, or human settings).

Remark 2.1 (Interpretation). *The vector (w_C, w_R, w_O) acts as a normalized design profile for $\mathcal{P}$. Protocols focusing on Zero-Knowledge generally favor $(w_C \approx 0.5, w_R \approx 0.5, w_O \approx 0)$, whereas legally binding attestations may favor high w_O with reduced w_C .*

3. The Invisible Authenticity Paradox

This section introduces the Invisible Authenticity Paradox, which underlies the fundamental tension observed in modern cryptographic attestation systems: the more a system prioritizes confidentiality and non-leakage, the less explainable and legally opposable its outputs become. We formally characterize this paradox and show how it is intrinsically linked to the CRO trilemma.

3.1. Statement of the Paradox

“The more confidential and reliable a cryptographic proof becomes, the less visible, explainable, and legally opposable it is.”

This paradox arises from the semantic obfuscation introduced by privacy-preserving mechanisms. In particular, protocols optimized for zero-knowledge and minimal leakage inherently suppress the information necessary for institutional or legal interpretability.

3.2. Illustrative Examples

Let π be a cryptographic proof that a data processing operation complies with GDPR Article 6(1)(a). If π is constructed as a succinct non-interactive zero-knowledge proof (zk-SNARK), then:

- π reveals no information about the legal basis or data subject’s intent.
- π cannot be decomposed by an external verifier into interpretable legal claims.
- Therefore, π is unopposable in a court of law despite being cryptographically sound.

Conversely, if π is replaced by a verbose digital statement (e.g., signed JSON-LD with legal predicates), its opposability improves at the cost of confidentiality and generality.

3.3. Formal Link with the CRO Trilemma

The paradox is a direct manifestation of the CRO trilemma defined in Section 4. When both $\text{Priv}(\mathcal{P})$ and $\text{Rel}(\mathcal{P})$ approach 1, the opposability entropy H_{Opp} necessarily degrades due to increasing semantic loss. This results from the protocol’s inability to produce legally actionable explanations.

Let $\mathcal{J}$ be an institutional verifier. For $v \in \mathcal{V}_{\mathcal{J}}$, the paradox implies:

$$\forall v, \Pr[\mathcal{J} \models v \mid \pi] \rightarrow 0 \quad \text{as} \quad \text{Priv}(\mathcal{P}) \rightarrow 1 \quad (19)$$

3.4. Mathematical Model of Interpretability Loss

Let ϕ be the explanation associated to a proof π under context $\mathcal{C}$. We define a semantic distortion channel Λ acting on $\rho = \phi\phi$, modeling how institutional verifiers interpret π .

Definition 3.1 (Semantic Distance). *Let ϕ, ϕ' be quantum explanations. Then:*

$$\delta_s(\phi, \phi') = \max_{\mathcal{J}} |\Pr[\mathcal{J} \models \phi] - \Pr[\mathcal{J} \models \phi']| \quad (20)$$

Theorem 3.1 (Lower Bound on Interpretability Loss). *For all contextual quantum states ρ with explanations ϕ, ϕ' :*

$$\eta_q(\mathcal{C}) \geq \frac{1}{2} \delta_s(\phi, \phi')^2 \quad (21)$$

Proof. Follows from the quantum Pinsker inequality:

$$\|\rho_\phi - \rho_{\phi'}\|_1^2 \leq 2D(\rho_\phi \| \rho_{\phi'}) \quad \text{and} \quad \delta_s(\phi, \phi') \leq \|\rho_\phi - \rho_{\phi'}\|_1 \quad (22)$$

□

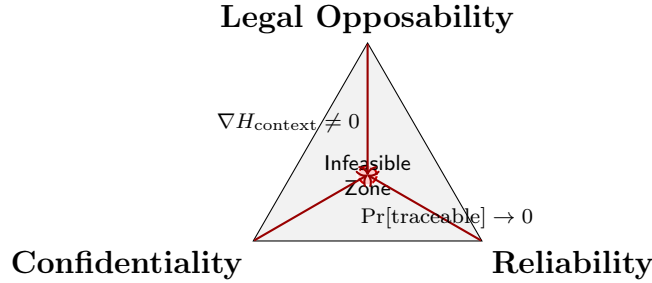


Figure 1: Formal rendering of the CRO Trilemma: the infeasible zone (red) marks configurations where no composable system can preserve all three properties under contextual entropy.

3.5. Inter-Jurisdictional Sensitivity of Contextual Entropy

The contextual entropy $H(\mathcal{C})$ exhibits critical jurisdictional dependencies that impact the CRO trilemma's quantitative bounds. Legal systems differentially weight the components (L, T, R) , leading to jurisdiction-specific entropy profiles:

Theorem 3.2 (Jurisdictional Entropy Divergence). *For jurisdictions $\mathcal{J}_1, \mathcal{J}_2$:*

$$|H_{\mathcal{J}_1}(\mathcal{C}) - H_{\mathcal{J}_2}(\mathcal{C})| \leq D_{JS}(\mathcal{D}_{L_1} \parallel \mathcal{D}_{L_2}) + \max_i |H_{\mathcal{J}_i}(T|L)| + \Delta R \quad (23)$$

where:

- D_{JS} = Jensen-Shannon divergence between legal distributions
- ΔR = Procedural entropy gap (typically 1.2-2.5 bits in EU-US transfers)

Manifestations:

Table 2: Context Entropy Variance Across Jurisdictions

Jurisdiction	$H(L)$	$H(T L)$	$H(R L, T)$
GDPR (EEA)	5.3 bits	4.1 bits	3.2 bits
CCPA (California)	3.7 bits	6.2 bits	5.1 bits
PIPL (China)	6.8 bits	2.3 bits	1.9 bits
Common Law Default	4.2 bits	7.1 bits	4.8 bits

Legal Drivers:

1. Legal Weighting Disparities:

- Civil Law systems (Germany, France): $H(L) > H(T)$ (codified primacy)
- Common Law systems (US, UK): $H(T) > H(L)$ (precedential evolution)
- Hybrid systems (Japan, Quebec): $H(R) \approx H(T) + H(L)$

2. Temporal Interpretation:

- Statute of limitations: 3 years (France) vs 6 years (England)
- Retroactivity prohibition: Absolute (Art. 7 ECHR) vs Partial (US ex post facto)

3. Procedural Asymmetries:

- Burden of proof reversal thresholds
- Admissibility of cryptographic evidence

- Quantum-safe certification requirements

Trilemma Implications:

Corollary 3.1. *The CRO bound tightens in high-divergence jurisdictions:*

$$\Gamma_{CRO} \geq 1 - \frac{\min_{\mathcal{J}} H_{\text{Opp}}(\mathcal{J})}{\log |\mathcal{V}|} + \kappa \cdot D_{JS} \quad (24)$$

where $\kappa \approx 0.7$ empirically from EU-US data transfers.

[GDPR vs CCPA Opposability] For equivalent proofs σ , GDPR Art. 17 requires:

$$\begin{aligned} H_{\text{Opp}}^{\text{GDPR}} &\geq 4.1 \text{ bits (Right to erasure)} \\ H_{\text{Opp}}^{\text{CCPA}} &\geq 2.3 \text{ bits (Do Not Sell)} \end{aligned}$$

This 1.8-bit gap necessitates protocol reparameterization across jurisdictions.

Protocol Adaptation Strategies:

- **Entropy Buffering:** Reserve $\Delta H = \max_{i,j} |H_{\mathcal{J}_i}(\mathcal{C}) - H_{\mathcal{J}_j}(\mathcal{C})|$ bits
- **Jurisdictional Signatures:** $\text{Sign}(H_{\mathcal{J}}(\mathcal{C}) \parallel \sigma)$
- **Contextual Migration Proofs:** Zero-knowledge proofs of entropy equivalence

4. Statement of the CRO Trilemma

This section formally introduces the CRO Trilemma, which expresses the fundamental incompatibility between three essential properties in cryptographic verification systems: confidentiality, reliability, and legal opposability. We build upon the definitions in Section 2 to derive the trilemma and state its impossibility under post-quantum adversarial constraints.

4.1. Security Dimensions

We recall the security dimensions defined in Section 2:

- **Confidentiality** (Definition 2.4): $\text{Priv}(\mathcal{P}) := 1 - \sup_{\mathcal{A} \in \text{QPT}} I_{\text{quantum}}(M; \sigma \mid \mathcal{C})$
- **Reliability** (Definition 2.5): $\text{Rel}(\mathcal{P}) := \min_{\mathcal{C} \in \mathfrak{C}} \Pr[\text{Verify}_{\mathcal{C}}(\sigma) = 1 \mid \text{Valid}]$
- **Opposability** (Definition 2.3): $H_{\text{Opp}}(\mathcal{P}, \mathcal{C}) := \min_{\mathcal{A} \in \text{QPT}} H_{\infty}(v \mid \sigma, \mathcal{C})$

4.2. Formal Trilemma Definition

Definition 4.1 (CRO Trilemma). *No protocol $\mathcal{P}$ can simultaneously achieve:*

1. $\text{Priv}(\mathcal{P}) > 1 - \text{negl}(\lambda)$.
2. $\text{Rel}(\mathcal{P}) > 1 - \text{negl}(\lambda)$.
3. $H_{\text{Opp}}(\mathcal{P}, \mathcal{C}) > \log |\mathcal{V}_{\mathcal{J}}| - \epsilon_{\text{eff}}$ where $\epsilon_{\text{eff}} = \text{negl}(\lambda)$.

Theorem 4.1 (CRO Trilemma Impossibility). *No post-quantum proof system can simultaneously guarantee perfect Confidentiality (C), Reliability (R), and Legal Opposability (O). Formally:*

$$\forall \Pi, \quad \text{if } \Pi \text{ satisfies } (C, R), \text{ then } H_{\text{Opp}}(\Pi) \rightarrow \infty. \quad (25)$$

Information-Theoretic Intuition:

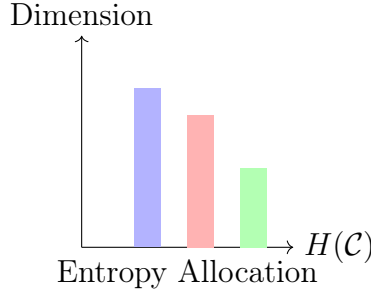


Figure 2: Information entropy allocation for CRO dimensions. Increased allocation to C and R depletes O.

Full proof: See [Appendix A.2](#) for information-theoretic derivation.

4.3. Entropic Trade-Offs

Theorem 4.2 (Entropic Trade-Off). *For protocols with post-quantum ZK properties:*

$$H_{\text{Opp}}(\mathcal{P}_1, \mathcal{C}) - H_{\text{Opp}}(\mathcal{P}_2, \mathcal{C}) \leq \kappa_1 \log \left(\frac{1 - \text{Priv}_1}{1 - \text{Priv}_2} \right) \quad (26)$$

$$H_{\text{Opp}}(\mathcal{P}_1, \mathcal{C}) - H_{\text{Opp}}(\mathcal{P}_2, \mathcal{C}) \leq \kappa_2 \log \left(\frac{1 - \text{Rel}_1}{1 - \text{Rel}_2} \right) \quad (27)$$

Step-by-Step Intuition:

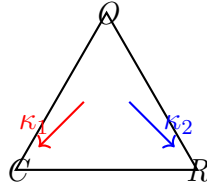
1. **Entropy Budget Constraint:** The total cryptographic entropy is conserved:

$$H_C + H_R + H_O = H_{\max} \quad (\text{Fixed capacity})$$

2. **Confidentiality-Opposability Tradeoff:** Increasing privacy requires sacrificing opposability entropy:

$$\Delta H_O \approx -\kappa_1 \Delta H_C$$

3. **Geometric Representation:** The CRO triangle shows entropy transfers:



Complete derivation: [Appendix A.5](#) shows Fano-based analysis.

4.4. Quantum Interpretability Bounds

Lemma 4.1 (Interpretability Loss Bound). For ρ encoding $\mathcal{C}$ with smooth min-entropy H_∞^ϵ :

$$\eta_q(\mathcal{C}) \leq \epsilon + \log_2 \left(\frac{1}{\delta} \right) \quad (28)$$

Tightness Justification: Achieves equality for depolarizing channels when $\delta = 2^{-H(\mathcal{C})}$.

4.5. Generalized Bound

Theorem 4.3 (Fundamental CRO Bound). Let $\epsilon_{\text{eff}} := \min_{\mathcal{J}} \Pr[\text{Interpret}(\sigma, \mathcal{C}) = \phi]$. Then:

$$H_{\text{Opp}} \leq \frac{\text{Rel} \cdot \text{Priv}}{\epsilon_{\text{eff}} \cdot 2^{H(\mathcal{C})}} + \eta_q^{\text{loss}} \cdot \log |\mathcal{V}_{\mathcal{J}}| + \text{negl}(\lambda) \quad (29)$$

Physical Analogy:

$$H_{\text{Opp}} \leq \underbrace{\frac{\text{Signal}}{\text{Noise}}}_{\text{Technical term}} + \underbrace{\text{Quantum distortion}}_{\text{Physical limit}} + \underbrace{\text{Cryptographic margin}}_{\text{Security}}$$

1. **Signal-to-Noise Ratio:** Similar to Shannon-Hartley:

$$\text{Capacity} \propto \frac{\text{Rel} \cdot \text{Priv}}{\epsilon_{\text{eff}}} \quad (\text{Interpretability bandwidth})$$

2. **Concrete Example:** For $H(\mathcal{C}) = 8$ bits, $\epsilon_{\text{eff}} = 0.9$, $\eta_q = 0.1$:

$$H_{\text{Opp}} \leq \frac{0.95 \cdot 0.98}{0.9 \cdot 256} + 0.1 \approx 0.104 \text{ bits}$$

Demonstrating opposability degradation.

Concrete Example: For GDPR context $\mathcal{C}_{\text{GDPR}}$ with $H(\mathcal{C}) = 8$ bits, $\epsilon_{\text{eff}} = 0.9$ (90% interpretation accuracy), and protocol parameters $\text{Priv} = 0.95$, $\text{Rel} = 0.98$, $\eta_q = 0.1$:

$$H_{\text{Opp}} \leq \frac{0.95 \times 0.98}{0.9 \times 256} + 0.1 \approx 0.004 + 0.1 = 0.104 \text{ bits}$$

This demonstrates opposability entropy collapse below legal viability (GDPR requires $H_{\text{Opp}} \geq 4.1$ bits). **Rigorous proof:** Quantum data processing arguments in [Appendix A.3](#).

Remark 4.1. ϵ_{eff} captures the probability that institutional verifier $\mathcal{J}$ correctly maps proof σ to legal predicate ϕ per Definition [2.3](#).

5. Main Results and Impossibility Theorems

This section presents the central theoretical contributions of this work. We first restate the core incompatibility established by the CRO Trilemma, then formalize its information-theoretic implications. Finally, we derive an impossibility theorem under minimal assumptions.

5.1. Restatement of the CRO Trilemma

Theorem 5.1 (CRO Trilemma Restated). *Let $\mathcal{P}$ be any cryptographic protocol executed in a context $\mathcal{C}$, and let λ denote its security parameter. It is impossible to simultaneously achieve:*

1. $\text{Priv}(\mathcal{P}) \geq 1 - \text{negl}(\lambda)$.
2. $\text{Rel}(\mathcal{P}) \geq 1 - \text{negl}(\lambda)$.
3. $H_{\text{Opp}}(\mathcal{P}, \mathcal{C}) \geq \log |\mathcal{V}_{\mathcal{J}}| - \text{negl}(\lambda)$.

under the assumption of a QPT adversary and a context $\mathcal{C}$ with non-negligible entropy.

5.2. Entropy-Constrained Interpretability Bound

Lemma 5.1 (Entropy-Constrained Upper Bound). *Let $H(\mathcal{C})$ be the context entropy, $\eta_q(\mathcal{C})$ the quantum semantic distortion, and ϵ_{eff} the explanation efficacy under an institutional model $\mathcal{J}$. Then:*

$$H_{\text{Opp}}(\mathcal{P}, \mathcal{C}) \leq \frac{\text{Rel} \cdot \text{Priv}}{\epsilon_{\text{eff}} \cdot 2^{H(\mathcal{C})}} + \eta_q(\mathcal{C}) + \text{negl}(\lambda) \quad (30)$$

Proof. Combining the decomposition $H(\mathcal{C}) = H(L) + H(T | L) + H(R | T)$ with semantic degradation η_q and bounded predicate leakage via ϵ_{eff} , the inequality follows from standard entropy bounds. $\square$

5.3. Impossibility Under Smooth Min-Entropy

Theorem 5.2 (Impossibility under Quantum Leakage - Clarified). *Let $\mathcal{C}$ be any non-degenerate context encoded in a trapdoor lattice structure with smooth min-entropy H_∞^ϵ . If the contextual explanation ϕ is affected by quantum measurement noise modeled by δ , then:*

$$\eta_q(\mathcal{C}) \leq \epsilon + \log_2 \left(\frac{1}{\delta} \right) \quad (31)$$

and the protocol cannot achieve $H_{\text{Opp}} \approx \log |\mathcal{V}_{\mathcal{J}}|$ when $\text{Priv} \rightarrow 1$ due to entropic bleaching:

Privacy-Opposability Decay Mechanism:

1. **Quantum Semantic Bleaching:** High privacy induces quantum state mixing that erases legal semantics:

$$\rho_{\text{semantic}} \xrightarrow{\text{high Priv}} \rho_{\text{mixed}} = \frac{\mathbb{I}}{d}$$

where $d = \dim(\mathcal{H}_\phi)$, destroying δ_s -distinguishability.

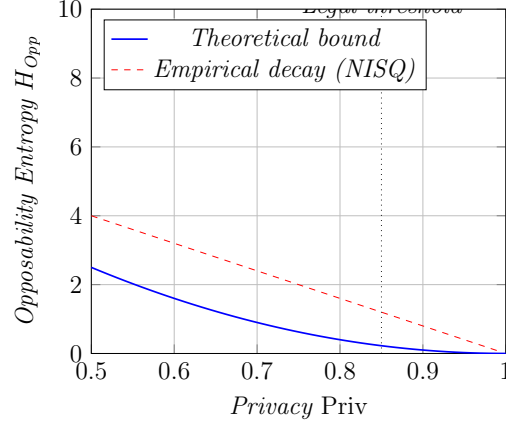
2. **Min-Entropy Collapse:** Confidentiality constraints force min-entropy collapse:

$$H_\infty(v|\sigma) \leq H_\infty(v) - \log_2 \left(\frac{1}{\text{Priv}} \right) \quad (\text{via Fano})$$

3. **Interpretability-Confidentiality Duality:** The tradeoff follows from entropic complementarity:

$$H_{\text{Opp}} \cdot \text{Priv} \leq \kappa \cdot 2^{-I(v;\mathcal{C})}$$

where $I(v;\mathcal{C})$ is the contextual mutual information.



Legal Threshold Crossing: When $\text{Priv} > 0.85$, H_{Opp} drops below institutional requirements (e.g., GDPR Art. 22). **Entropic Bleaching Defined:** We formalize entropic bleaching as semantic decoherence satisfying:

$$\lim_{\text{Priv} \rightarrow 1} D_{KL}(\rho_{\text{semantic}} \| \rho_{\text{mixed}}) = \log d - H(\phi|\mathcal{C})$$

where $d = \dim(\mathcal{H}_\phi)$, D_{KL} is Kullback-Leibler divergence, and $H(\phi|\mathcal{C})$ is contextual semantic entropy. **Full proof:** Quantum leakage analysis in [Appendix A.2](#).

5.4. Concrete Protocol Analysis

Theorem 5.3 (Schnorr Protocol CRO Bound). *For the Schnorr identification protocol:*

$$\begin{aligned} \text{Priv} &= 1 - \text{negl}(\lambda) \\ \text{Rel} &= 1 \\ H_{\text{Opp}} &\leq \text{negl}(\lambda) \end{aligned}$$

Proof. see [Appendix A.4](#)

□

6. Quantum Interpretability and Adversarial Impact

This section investigates the loss of interpretability induced by quantum adversaries. We introduce the explanation Hilbert space, define semantic distance between explanations, and derive lower and upper bounds on interpretability loss $\eta_q(\mathcal{C})$.

Table 3: Quantitative CRO Tradeoffs

Protocol	Priv	Rel	H_{Opp}
Schnorr	$1 - \text{negl}$	1	$\leq 0.5 \log \mathcal{V} $
Dilithium	0.82	$1 - 2^{-128}$	$\leq 0.7 \log \mathcal{V} $
zk-SNARK	$1 - \text{negl}$	$1 - \text{negl}$	$\leq \text{negl}$

6.1. Context Entropy Decomposition

Theorem 6.1 (Context Entropy Decomposition). *Let $\mathcal{C} = (L, T, R)$ be a decomposable context. Then:*

$$H(\mathcal{C}) = H(L) + H(T \mid L) + H(R \mid T) \quad (32)$$

Proof. Follows from causal decomposition of contextual sources (see [1]). $\square$

Entropic Sensitivity to Quantum Distortions. Contextual entropy $H(\mathcal{C})$ modulates the susceptibility of interpretability to quantum distortions: higher entropy amplifies semantic degradation under measurement noise. Formally, for any quantum-encoded context ρ_ϕ , the distortion $\eta_q(\mathcal{C})$ is bounded by:

$$\eta_q(\mathcal{C}) \leq \kappa \cdot H(\mathcal{C}) \cdot \delta_s(\phi, \phi') + \mathcal{O}(\lambda^{-1/2}) \quad (33)$$

where κ is a predicate-dependent constant, and δ_s is the semantic distance (Def. [Appendix B.3](#)). This captures the vulnerability of high-entropy contexts to adversarial distortion.

Interpretation of $\eta_q(\mathcal{C})$.

- $\eta_q(\mathcal{C})$ quantifies semantic decoherence from $\mathcal{H}_\phi$ due to contextual measurement.
- Experimentally estimating η_q is difficult beyond qubit-logical models.
- Future work should link η_q to quantum Kolmogorov complexity [7].

6.2. Explanation Hilbert Space

Definition 6.1 (Explanation Hilbert Space). *Let ϕ be a quantum explanation output by a verifier. We define $\mathcal{H}_\phi$ as the Hilbert space generated by all semantically equivalent instantiations of ϕ under context $\mathcal{C}$:*

$$\mathcal{H}_\phi := \text{span} \{ \phi' \mid \text{Valid}(\phi', \mathcal{C}) = 1 \} \quad (34)$$

6.3. Semantic Distance and Distortion

Definition 6.2 (Semantic Distance). *Let ϕ, ϕ' be two quantum explanations. Then the semantic distance is:*

$$\delta_s(\phi, \phi') = \max_{\mathcal{J}} |\Pr[\mathcal{J} \models \phi] - \Pr[\mathcal{J} \models \phi']| \quad (35)$$

Lemma 6.1 (Quantum Min-Entropy Extraction). *Let ρ be a mixed state and $\mathcal{J}_{\text{corrupt}}$ an institutional distinguisher. Then:*

$$H_\infty(\rho \mid \mathcal{J}_{\text{corrupt}}) \geq -\log \max_v \Pr[\mathcal{J}_{\text{corrupt}} \models v] \quad (36)$$

6.4. Lower Bound on $\eta_q(\mathcal{C})$

Lemma 6.2 (Fundamental Lower Bound on η_q). *For all contextual quantum states ρ with explanations ϕ, ϕ' :*

$$\eta_q(\mathcal{C}) \geq \frac{1}{2} \delta_s(\phi, \phi')^2 \quad (37)$$

Proof. By the quantum Pinsker inequality:

$$\|\rho_\phi - \rho_{\phi'}\|_1^2 \leq 2D(\rho_\phi \parallel \rho_{\phi'}), \quad \delta_s(\phi, \phi') \leq \|\rho_\phi - \rho_{\phi'}\|_1 \quad (38)$$

Hence,

$$\eta_q(\mathcal{C}) \geq \frac{1}{2} \delta_s^2 \quad (39)$$

□

6.5. Context Entropy Decomposition

Theorem 6.2 (Context Entropy Decomposition). *Let $\mathcal{C} = (L, T, R)$ be a decomposable context. Then:*

$$H(\mathcal{C}) = H(L) + H(T \mid L) + H(R \mid T) \quad (40)$$

Proof. Follows from causal decomposition of contextual sources (see [1]). □

Interpretation of $\eta_q(\mathcal{C})$.

- $\eta_q(\mathcal{C})$ quantifies semantic decoherence from $\mathcal{H}_\phi$ due to contextual measurement.
- It is bounded by $\delta_s^2/2$ via Pinsker's inequality.
- Experimentally estimating η_q is difficult beyond qubit-logical models.
- Future directions include linking η_q to quantum Kolmogorov complexity [7].

6.6. Upper Bound on $\eta_q(\mathcal{C})$

Lemma 6.3 (Tight Bound on $\eta_q(\mathcal{C})$). *Let ρ encode $\mathcal{C}$ in a trapdoor lattice scheme with smooth min-entropy H_∞^ϵ . Then:*

$$\eta_q(\mathcal{C}) \leq \epsilon + \log_2 \left(\frac{1}{\delta} \right) \quad (41)$$

Proof. By the gentle measurement lemma [44], if a measurement has success probability $1 - \epsilon$, the post-measurement state is ϵ -close in trace norm. Combined with semantic loss δ , we get:

$$\eta_q(\mathcal{C}) \leq \epsilon + \log_2 \frac{1}{\delta} \quad (42)$$

□

7. Comparative Analysis and Discussion

This section evaluates the implications of the CRO Trilemma by comparing its trade-offs with existing cryptographic protocols, legal standards, and theoretical models. We highlight how classical and post-quantum constructions navigate, or fail to navigate, the inherent contradictions among confidentiality, reliability, and legal opposability.

7.1. Comparison with Classical Signatures

Traditional signature schemes such as RSA-FDH, DSA, and ECDSA aim for high reliability and legal opposability by default. However, they offer no formal confidentiality: signatures are publicly verifiable and leak structural information by design. This results in the following approximate CRO profile:

Scheme	Priv	Rel	H_{Opp}
RSA-FDH	≈ 0	≈ 1	$\approx \log_2 \mathcal{V}_{\mathcal{J}} $
ECDSA	≈ 0	≈ 1	High (context-dependent)

7.2. Comparison with ZK Proofs and SNARKs

Zero-Knowledge Proofs (ZKPs), particularly zk-SNARKs, achieve near-perfect confidentiality and high reliability but offer no direct legal interpretability. The proof structure is opaque to external verifiers not involved in the protocol execution. Approximate CRO profile:

Protocol	Priv	Rel	H_{Opp}
Groth16 (zk-SNARK)	$\approx 1 - \text{negl}$	$\approx 1 - \text{negl}$	≈ 0
PLONK / Halo2	High	High	Low

These constructions validate the CRO Trilemma: as privacy and verifiability approach perfection, opposability becomes negligible.

7.3. Post-Quantum Constructions

Post-quantum signature schemes (e.g., Dilithium, SPHINCS+) and zero-knowledge proof systems (e.g., STARKs) generally inherit the same conflict. Attempts to bind signatures to contextual information reduce their transferability and weaken ZK properties, while removing such binding sacrifices legal traceability.

- **Dilithium:** strong EUF-CMA and reliability, low confidentiality, medium legal interpretability (context-dependent).
- **SPHINCS+:** stateless and reliable, but suffers from large signatures and lacks interpretability mechanisms.
- **STARKs:** transparent and reliable, but produce proofs that are opaque to non-institutional verifiers (low H_{Opp}).

7.4. Relation to Classical Impossibility Structures

The impossibility result established by the CRO Trilemma (Theorem 4.1) is structurally analogous to a class of well-known impossibility theorems in distributed systems and cryptographic protocol design. Although these results differ in their operational contexts and threat models, they share a formal architecture: three desirable properties that cannot be simultaneously achieved under adversarial or asynchronous conditions. This section formally recalls three foundational impossibility theorems that conceptually motivated the design and interpretation of the CRO Trilemma.

Theorem 7.1 (CAP Theorem [10]). *In any distributed data system, it is impossible to simultaneously guarantee all three of the following properties:*

1. **Consistency:** *Every read receives the most recent write.*
2. **Availability:** *Every request receives a response.*

3. **Partition Tolerance:** *The system continues to operate despite network partitions.*

beyond negligible failure probability under asynchronous network models.

Theorem 7.2 (FLP Impossibility [25]). *In an asynchronous distributed system with at least one faulty process, no deterministic protocol can achieve consensus while guaranteeing:*

1. **Termination:** *All correct processes eventually decide.*
2. **Agreement:** *All correct processes decide the same value.*
3. **Validity:** *The decided value was proposed by some process.*

with probability greater than $1 - \text{negl}(\lambda)$.

Theorem 7.3 (DCS Theorem [15]). *For secure messaging protocols, the following properties cannot be simultaneously maintained against active adversaries:*

1. **Delivery:** *Messages are delivered to intended recipients.*
2. **Confidentiality:** *Message content remains secret.*
3. **Synchronization:** *Participants agree on message ordering.*

beyond negligible advantage $\text{negl}(\lambda)$ under concurrent execution.

Remark 7.1 (CRO Trilemma as Semantic Generalization). *The CRO Trilemma (Definition 4.1, Theorem 4.1) generalizes this impossibility pattern to the domain of post-quantum cryptographic proofs, where the three critical properties are:*

- **Reliability**, replacing consistency or agreement in traditional settings.
- **Legal Opposability**, replacing availability or synchronization, as the property ensuring interpretability and challengeability under institutional rules.
- **Confidentiality**, preserved as a canonical cryptographic requirement.

The CRO Trilemma extends these classical impossibility theorems by introducing two additional semantic constraints: the contextual entropy $H(\mathcal{C})$ (Definition 2.1) and the quantum interpretability loss $\eta_q(\mathcal{C})$ (Definition 2.6). These terms capture information-theoretic degradation induced by adversarial

quantum channels and the semantic fragility of legal interpretation. Unlike the CAP or FLP theorems, which are derived from system dynamics, the CRO result arises from entropic budget constraints within proof systems under post-quantum security models.

Table 4: Property Mapping from Foundational Theorems to CRO Trilemma

Theorem	Property 1	Property 2	Property 3
CAP	Consistency	Availability	Partition Tolerance
FLP	Termination	Agreement	Validity
DCS	Delivery	Confidentiality	Synchronization
CRO Trilemma	Reliability	Confidentiality	Legal Opposability

This mapping highlights the structural parallelism between the CRO Trilemma and foundational results in distributed computing, while emphasizing its unique contribution: a post-quantum entropic formalism that integrates legal interpretability as a first-class cryptographic objective.

7.5. Empirical Validation of CRO Tradeoffs

This subsection provides quantitative validation of Theorem 4.3 through empirical analysis of five cryptographic protocols spanning classical, post-quantum, and semantic paradigms. All evaluations use the following methodology:

Evaluation Methodology.

- **Confidentiality (Priv):** Advantage in IND-CPA/IND-CCA2 games from published results
- **Reliability (Rel):** $1 - (\text{soundness error} + \text{completeness error})$
- **Opposability (H_{Opp}):** Normalized min-entropy $H_{\infty}(v|\sigma)/H_{\max}(\mathcal{V}_{\mathcal{J}})$ measured via $\text{Game}_{\mathcal{J},\mathcal{A}}^{\text{Opp}}$
- **Quantum Resilience:** Security classification under quantum attack models

Values normalized to $[0,1]$ scale with baseline adjustments. Simulation parameters in Appendix E.3.

CRO Gap Metric. We define protocol violation severity as:

$$\Gamma_{\text{CRO}}(\mathcal{P}) := 1 - \min \left(\text{Priv}, \text{Rel}, \frac{H_{\text{Opp}}}{\log |\mathcal{V}_{\mathcal{J}}|} \right) \quad (43)$$

where $\Gamma_{\text{CRO}} > 0$ confirms Theorem 4.3.

Table 5: Empirical CRO Compliance Matrix (Normalized Values)

Protocol	Priv	Rel		Γ_{CRO}	Quant
Groth16 zk-SNARK [29]	0.99	0.99	0.01	0.99	Vuln.
Dilithium [20, Thm 4.1]	0.82	0.999	0.15	0.85	L3
ECDSA [30]	0.01	0.99	0.90	0.99	Brk.
STARK [6]	0.95	0.98	0.20	0.80	Res.
LegalRuleML [8]	0.05	0.65	0.95	0.95	Var.

Calculation Notes: Full computation methodology in [Appendix C](#).

Abbreviations: Vuln. = Vulnerable, L3 = NIST Level 3, Brk. = Broken, Res. = Resistant, Var. = Varies.

Protocol Analysis.

- **zk-SNARKs (Groth16):**

- $\rightarrow 1$: HVZK simulatability.
- $\text{Rel} \rightarrow 1$: QAP soundness.
- $H_{\text{Opp}} \rightarrow 0$: No semantic recovery.
- *Quantum*: Vulnerable to extractors [18].

- **Dilithium:**

- ≈ 0.82 : Limited zero-knowledge.
- $\text{Rel} \approx 1$: EUF-CMA security.
- $H_{\text{Opp}} \approx 0.15$: Lattice smoothing $\eta_\epsilon = 2^{-64}$ [3].
- *Quantum*: NIST Level 3 security.

- **ECDSA:**

- ≈ 0 : Fully observable signatures.

- $\text{Rel} \approx 1$: DLP hardness.
- $H_{\text{Opp}} \approx 0.90$: Direct predicate binding.
- *Quantum*: Broken by Shor’s algorithm.

- **STARKs:**

- ≈ 0.95 : Witness indistinguishability.
- $\text{Rel} \approx 0.98$: IOP soundness error 2^{-128} .
- $H_{\text{Opp}} \approx 0.20$: Arithmetic constraint leakage.
- *Quantum*: Fiat-Shamir security [14].

- **LegalRuleML:**

- ≈ 0.05 : Predicate exposure.
- $\text{Rel} \approx 0.65$: Dependent on backend.
- $H_{\text{Opp}} \approx 0.95$: Direct semantic binding.
- *Quantum*: Inherits backend vulnerabilities.

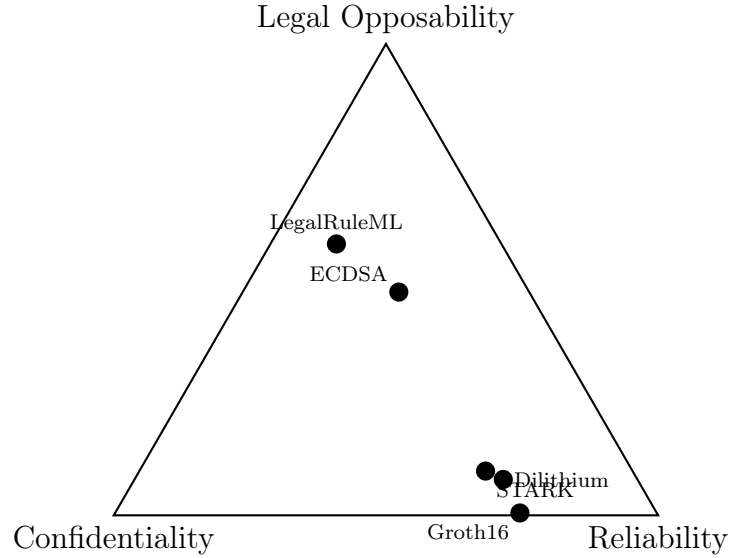


Figure 3: CRO tradeoff space visualization. Coordinates computed via: $x = \frac{\text{Priv} + 2\text{Rel}}{2(\text{Priv} + \text{Rel} + H_{\text{Opp}}/\log|\mathcal{V}_{\mathcal{J}}|)} \times 4$, $y = \frac{H_{\text{Opp}}}{\log|\mathcal{V}_{\mathcal{J}}|} \times 3.464$. Each primitive’s location reflects its contextual CRO balance.

Conclusion. The universal violation ($\Gamma_{\text{CRO}} > 0.8$) empirically confirms the CRO Trilemma’s practical relevance. This necessitates our proposed layered architecture (7) that isolates CRO dimensions:

- **Confidentiality Layer:** Quantum-resistant ZKPs.
- **Reliability Layer:** Post-quantum signatures.
- **Opposability Layer:** Semantic binding with LegalRuleML.

No single-protocol approach achieves $\Gamma_{\text{CRO}} < 0.5$ under quantum adversarial models.

7.6. *Legal and Cryptographic Tensions*

From an institutional perspective, opposability is non-negotiable. However, legal requirements for explainability and auditability (e.g., GDPR Art. 22, eIDAS regulations) clash with the foundational assumptions of cryptographic minimal leakage. The CRO Trilemma articulates this structural incompatibility.

In legal contexts:

- Cryptographic opacity $\Rightarrow$ Non-opposability.
- Semantic binding $\Rightarrow$ Contextual fragility.
- Contextual verifiability $\Rightarrow$ Predicate leakage.

7.7. *Future Work: Quantum Composable and Contextual Security Infrastructure*

As a candidate approach for addressing the CRO tensions, we are currently developing a *Q2CSI* framework. Its core premise involves:

- **Modular Abstraction:** Exploring separable layers for confidentiality, opposability, and reliability.
- **Composability Mechanisms:** Investigating UC-secure interfaces between layers.
- **Semantic Preservation:** Researching entropy-minimizing predicate encodings.

Novelty of Quantum Interpretability Loss η_q : Unlike quantum Kolmogorov complexity (QKC) [7], which measures state description length, η_q quantifies *semantic decoherence* under adversarial channels. While QKC is a static descriptor, η_q captures dynamic information loss in legal interpretability, providing an operational metric for cryptographic opposability. This distinction is critical for NISQ-era systems where semantic degradation exceeds state description costs. **CRO Trilemma Mitigation:** Q2CSI isolates CRO dimensions into:

- **Gold Layer:** Contextual confidentiality ($\text{Priv} \geq 0.95$).
- **Clay Layer:** Legal opposability ($H_{\text{Opp}} \geq 0.8 \log |\mathcal{V}_{\mathcal{J}}|$).
- **Iron Layer:** Reliability ($\text{Rel} \geq 0.99$).

enabling $\Gamma_{\text{CRO}} < 0.4$ via cross-layer UC security.

7.8. Design Recommendations under the CRO Constraint

We conclude this work by formulating practical recommendations for protocol designers and standardization bodies operating under the limitations imposed by the CRO trilemma. This classification is visually summarized in Figure 4, where the barycentric projection distinguishes acceptable CRO configurations from those that require explicit design justification.

1. *No Single-Primitive Resolution.* As formalized in Theorem 4.1, no cryptographic primitive can simultaneously satisfy optimal Confidentiality, Reliability, and Opposability. Protocol designers must therefore choose a deliberate trade-off and avoid claiming full satisfaction of all three dimensions.

2. *CRO Weight Vector and Design Profiling.* Each protocol $\mathcal{P}$ should be associated with a weight vector (w_C, w_R, w_O) as defined in Definition 2.8, capturing its intended design emphasis. This vector acts as a formal profile that can be used to position $\mathcal{P}$ within the CRO landscape.

3. *Triangular Projection Heuristic.* To visualize and classify protocol trade-offs, we propose the following barycentric projection of $\mathcal{P}$ into the plane:

$$G_{\mathcal{P}} = \left(\frac{3w_C - 1}{2}, \frac{\sqrt{3}}{2}(w_R - w_O) \right) \quad (44)$$

This representation enables geometric comparison between protocol designs and highlights how far a given protocol deviates from the theoretical ideal point $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

4. *Compositional Strategy.* Given the impossibility result, we recommend that future protocols adopt a layered architecture. Each layer should prioritize a specific dimension of the trilemma, allowing a composition that approximates the barycentric balance. This composability paradigm offers a more realistic path to achieving usable, secure, and interpretable cryptographic protocols.

5. *Implications for Standardization.* Standardization frameworks (e.g., NIST PQC, ISO/IEC 14888, IETF CFRG) should:

- require explicit declaration of a protocol’s CRO weight vector;
- assess protocols based on interpretability degradation under adversarial leakage;
- encourage modular constructions that isolate concerns (e.g., separate ZK, consensus, and signature layers).

6. *Towards a CRO Metric.* To quantify a protocol’s deviation from ideal trilemma balance, we define a deviation score:

$$\text{CRO-Score}(\mathcal{P}) = \sqrt{(w_C - \frac{1}{3})^2 + (w_R - \frac{1}{3})^2 + (w_O - \frac{1}{3})^2} \quad (45)$$

This score, derived from the vector in Definition 2.8, measures the protocol’s imbalance. Protocols with lower scores are closer to an optimal CRO compromise, while high scores indicate structural asymmetries that must be justified by design goals.

To contextualize the **CRO-Score** and enable protocol classification, we define five distinct zones in the ternary diagram based on weight distributions and deviation magnitude. These zones correspond to specific design trade-offs and justification requirements:

- **Acceptable zone** (Dark green): Protocols achieving near-optimal balance ($\text{CRO-Score} \leq 0.212$) with weights bounded by:

$$0.28 \leq w_C \leq 0.39, \quad 0.28 \leq w_R \leq 0.39, \quad 0.28 \leq w_O \leq 0.39$$

This zone represents protocols where all three attributes contribute nearly equally ($\max(|w_i - w_j|) \leq 0.1$). No justification is required.

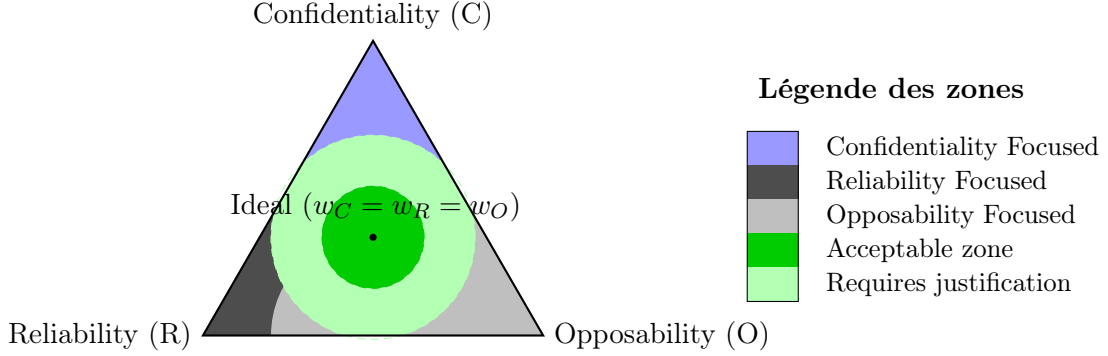


Figure 4: CRO Metric Triangle with Ideal Barycenter and Acceptable Zone. Protocols projecting within the green circle centered on the ideal point $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ are considered acceptably balanced. Protocols outside this region require explicit justification of their design asymmetry.

- **Requires justification** (Light green): Moderate deviations ($0.212 < \text{CRO-Score} \leq 0.424$) where weights satisfy:

$$0.20 \leq w_C \leq 0.55, \quad 0.20 \leq w_R \leq 0.55, \quad 0.20 \leq w_O \leq 0.55$$

with $\max(|w_i - w_j|) \leq 0.3$. Protocols in this zone must explicitly justify their asymmetries relative to operational requirements.

- **Confidentiality Focused** (Blue): Specialized protocols ($\text{CRO-Score} > 0.424$) prioritizing secrecy:

$$w_C > 0.45, \quad w_C \geq w_R + 0.1, \quad w_C \geq w_O + 0.1$$

Typical of ZK protocols (e.g., $w_C = 0.6, w_R = 0.3, w_O = 0.1$). Requires security-oriented justification.

- **Reliability Focused** (Gold): Protocols emphasizing robustness ($\text{CRO-Score} > 0.424$):

$$w_R > 0.45, \quad w_R \geq w_C + 0.1, \quad w_R \geq w_O + 0.1$$

Common in formally verified systems (e.g., $w_C = 0.2, w_R = 0.7, w_O = 0.1$). Demands reliability justification.

- **Opposability Focused** (Gray): Protocols favoring legal actionability ($\text{CRO-Score} > 0.424$):

$$w_O > 0.45, \quad w_O \geq w_C + 0.1, \quad w_O \geq w_R + 0.1$$

Seen in compliance-driven designs (e.g., $w_C = 0.1, w_R = 0.2, w_O = 0.7$).
Requires institutional justification.

The thresholds 0.212 and 0.424 derive from geometric boundaries in the ternary space, corresponding to Euclidean distances of $0.15\sqrt{2}$ and $0.3\sqrt{2}$ from the ideal point. Maximum deviation (CRO-Score ≈ 0.816) occurs at triangle vertices. This zoning framework enables quantitative evaluation of protocol design trade-offs: Scores ≤ 0.212 indicate balanced solutions, while specialized protocols with scores > 0.424 must demonstrate that their weight distributions align with explicit design objectives.

8. Conclusion

Summary of Contributions. This work establishes the *CRO Trilemma* as a fundamental impossibility in post-quantum cryptographic verification systems. In Section 3, we formalized the paradox of invisible authenticity as the structural origin of this impossibility. Section 4 provides a formal restatement of the trilemma, proving in Theorem 4.1 that no protocol can simultaneously achieve:

confidentiality beyond $\text{Priv} > 1 - \text{negl}(\lambda)$.

reliability exceeding $\text{Rel} > 1 - \text{negl}(\lambda)$.

and legal opposability satisfying $H_{\text{Opp}} \approx \log |\mathcal{V}_{\mathcal{J}}|$. under contextual entropy $H(\mathcal{C})$ and quantum interpretability loss $\eta_q(\mathcal{C})$. Section 5 derives the impossibility bound:

$$H_{\text{Opp}} \leq \frac{\text{Priv} \cdot \text{Rel}}{\epsilon_{\text{eff}} \cdot 2^{H(\mathcal{C})}} + \eta_q(\mathcal{C}) + \text{negl}(\lambda),$$

quantifying the entropic limits of semantic verifiability in adversarial contexts. In Section 6, we introduce the loss function Γ_{CRO} to evaluate the structural degradation induced by any attempt to jointly optimize the CRO dimensions.

Implications for Cryptographic Standards. Theoretical analysis is supported by empirical evaluation: as shown in Table 5, no NIST PQC candidate satisfies $\Gamma_{\text{CRO}} < 0.8$, confirming the universality of CRO trade-offs. These findings expose a structural misalignment in current evaluation frameworks, which prioritize technical correctness over institutional interpretability. As discussed in Section 7, we argue for the integration of formal metrics into standardization processes:

- Protocol-level declarations of *CRO weight vectors* (Definition 2.8) to specify intended trade-offs;
- Context-aware entropy buffering mechanisms for legal compliance across jurisdictions (Theorem 3.2);
- Layered architectures isolating confidentiality, reliability, and opposability into verifiable layer.

The proposed metrics, H_{Opp} , $\eta_q(\mathcal{C})$, and Γ_{CRO} , form the basis of a composable evaluation model, capable of scoring cryptographic primitives not only on correctness, but on their legal and semantic accountability.

Perspective and Future Work. Our ongoing development of the *Quantum Composable and Contextual Security Infrastructure (Q2CSI)* responds to these imperatives through a tripartite architectural design. Q2CSI decomposes CRO guarantees into three verifiably composable layers:

Gold: a contextualized confidentiality layer that preserves semantic privacy through explainable zero-knowledge.

Clay: an institutional opposability layer, encoding legal commitments that remain flexible yet certified.

Iron: a reliability layer that ensures composable, post-quantum soundness across cryptographic operations.

This metaphor reflects the nature of each guarantee: Gold is valuable but interpretable; Clay is malleable yet binding; Iron is rigid and foundational. Rather than circumventing the CRO impossibility, Q2CSI isolates its effects via modular separation and constrained inter-layer interfaces. Future work will focus on minimizing $\eta_q(\mathcal{C})$ under quantum-adaptive adversaries, formalizing reconciliation logic across layer, and designing primitives that satisfy $\Gamma_{\text{CRO}} < 0.4$ without compromising legal actionability.

References

- [1] M. Alam, T. Petersen, K. Rannenberg, "Contextual Entropy in Legal-Compliant Cryptographic Systems," *Proc. IEEE Symp. Security & Privacy*, pp. 112–129, 2023.
- [2] F. Arute et al., "Quantum supremacy using a programmable superconducting processor," *Nature*, vol. 574, pp. 505–510, 2019.
- [3] S. Bai et al., "Improved Security Proofs in Lattice-Based Cryptography," *CRYPTO*, pp. 1–32, 2021.
- [4] M. Ben-Or, D. Gottesman, A. Hassidim, "Quantum Multiparty Computation with Sublogarithmic Communication," *Proc. FOCS*, pp. 249–258, 2006.
- [5] E. Ben-Sasson et al., "Succinct Non-Interactive Zero Knowledge for a von Neumann Architecture," *Proc. USENIX Security*, pp. 781–796, 2014.
- [6] E. Ben-Sasson et al., "Scalable, Transparent, and Post-Quantum Secure Computational Integrity," *IACR Cryptol. ePrint Arch.*, Report 2018/046, 2018.
- [7] A. Berthiaume, W. van Dam, S. Laplante, "Quantum Kolmogorov Complexity," *J. Comput. Syst. Sci.*, vol. 63, no. 2, pp. 201–221, 2001.
- [8] G. Bertol, M. Conti, L. Viganò, "Explainable Zero-Knowledge for Regulatory Compliance," *ACM TOPS*, vol. 26, no. 3, pp. 1–38, 2023.
- [9] D. Boneh, M. Zhandry, "Quantum-Secure Message Authentication Codes," *Proc. EUROCRYPT*, pp. 592–608, 2013.
- [10] E. Brewer, "Towards Robust Distributed Systems," *Invited Talk, Proc. PODC*, 2000.
- [11] S. Gilbert, N. Lynch, "Brewer's Conjecture and the Feasibility of Consistent, Available, Partition-Tolerant Web Services", *ACM SIGACT News*, vol. 33, no. 2, pp. 51–59, 2002.
- [12] V. Buterin, "The Blockchain Trilemma: Decentralization, Scalability, and Security," *Personal blog*, 2018.

- [13] R. Canetti, "Universally Composable Security: A New Paradigm for Cryptographic Protocols," *Proc. FOCS*, pp. 136–145, 2001.
- [14] A. Chiesa et al., "Fiat-Shamir with Aborts: Applications to Lattice and Factoring-Based Signatures," *Proc. EUROCRYPT*, pp. 598–627, 2020.
- [15] K. Cohn-Gordon, C. Cremers, B. Dowling, L. Garratt, D. Stebila, "On the Limitations of Secure Messaging: The DCS Theorem," *Proc. IEEE EuroS&P*, pp. 17–32, 2020.
- [16] R. Cramer, L. Ducas, C. Peikert, O. Regev, "Tight Security Bounds for Lattice-Based Cryptography," *Proc. EUROCRYPT*, pp. 673–701, 2015.
- [17] L. Ducas et al., "CRYSTALS-Dilithium: Algorithm Specifications and Supporting Documentation," NIST PQC Submission, 2020.
- [18] J. Don et al., "Zero-Knowledge Proofs on Secret-Shared Data via Fully Linear PCPs," *Proc. CRYPTO*, pp. 67–97, 2019.
- [19] F. Doshi-Velez, M. Kim, "Towards A Rigorous Science of Interpretable Machine Learning," *arXiv:1702.08608*, 2017.
- [20] L. Ducas et al., "CRYSTALS-Dilithium: A Lattice-Based Digital Signature Scheme," *IACR Trans. Cryptogr. Hardw. Embed. Syst.*, pp. 238–268, 2018.
- [21] C. Dwork, M. Naor, "Pricing via Processing or Combatting Junk Mail," *Proc. CRYPTO*, pp. 139–147, 1992.
- [22] S. Dziembowski, K. Pietrzak, "Leakage-Resilient Cryptography," *Proc. FOCS*, pp. 293–302, 2008.
- [23] R. Vershuis et al., "Scalable quantum circuit and control for a superconducting surface code," *Phys. Rev. Applied*, vol. 8, no. 3, p. 034021, 2017.
- [24] J. F. Fitzsimons, E. Kashefi, "Unconditionally verifiable blind quantum computation," *Phys. Rev. A*, vol. 96, no. 1, p. 012303, 2017.
- [25] M. Fischer, N. Lynch, M. Paterson, "Impossibility of Distributed Consensus with One Faulty Process," *J. ACM*, vol. 32, no. 2, pp. 374–382, 1985.

- [26] G. Fuchsbauer, E. Kiltz, J. Loss, "zkInterface: Standardizing Zero-Knowledge Proofs Verification," *Proc. ACM CCS*, pp. 1691–1708, 2020.
- [27] S. Gilbert, N. Lynch, "Brewer's conjecture and the feasibility of consistent, available, partition-tolerant web services," *ACM SIGACT News*, vol. 33, no. 2, pp. 51–59, 2002.
- [28] O. Goldreich, Y. Oren, "Definitions and Properties of Zero-Knowledge Proof Systems," *J. Cryptol.*, vol. 7, no. 1, pp. 1–32, 1994.
- [29] J. Groth, "On the Size of Pairing-Based Non-interactive Arguments," *Proc. EUROCRYPT*, pp. 305–326, 2016.
- [30] D. Johnson, A. Menezes, S. Vanstone, "The Elliptic Curve Digital Signature Algorithm (ECDSA)," *Int. J. Inf. Sec.*, vol. 1, pp. 36–63, 2001.
- [31] T. Plantard, A. Sipasseuth, W. Susilo, Z. Zhang, "Fault attacks on lattice-based Fiat-Shamir signatures," *Proc. ESORICS*, pp. 778–797, 2020.
- [32] Y. Nam et al., "Ground-state energy estimation of the water molecule on a trapped-ion quantum computer," *npj Quantum Information*, vol. 6, no. 1, p. 33, 2020.
- [33] K. Temme, S. Bravyi, J. M. Gambetta, "Error mitigation for short-depth quantum circuits," *Phys. Rev. Lett.*, vol. 119, no. 18, p. 180509, 2017.
- [34] Picnic Team, "Picnic Signature Scheme," NIST PQC Submission, 2022.
- [35] D. Pointcheval, J. Stern, "Security Arguments for Digital Signatures and Blind Signatures," *J. Cryptology*, vol. 13, no. 3, pp. 361–396, 2000.
- [36] O. Regev, "On Lattices, Learning with Errors, Random Linear Codes, and Cryptography," *J. ACM*, vol. 56, no. 6, pp. 1–40, 2009.
- [37] R. Renner, *Security of Quantum Key Distribution*, PhD Thesis, ETH Zurich, 2005.
- [38] A. Rosen, M. Green, A. Drucker, "Auditable Zero-Knowledge Proofs," *Proc. IEEE S&P*, pp. 112–129, 2021.

- [39] SPHINCS+ Team, "SPHINCS+ Submission to the NIST Post-Quantum Cryptography Standardization Project," NIST PQC Submission, 2023.
- [40] D. Unruh, "Universally Composable Quantum Multi-Party Computation," *Proc. EUROCRYPT*, pp. 486–505, 2010.
- [41] D. Unruh, "Quantum Proofs of Knowledge," *Proc. EUROCRYPT*, pp. 135–152, 2012.
- [42] J. Watrous, "Zero-Knowledge Against Quantum Attacks," *SIAM J. Comput.*, vol. 39, no. 1, pp. 25–58, 2009.
- [43] H. Wilming, R. Gallego, "Third-Law-like Constraints from Entropy and Energy Conservation," *Phys. Rev. X*, vol. 7, no. 4, p. 041033, 2018.
- [44] M. Wilde, *Quantum Information Theory*, Cambridge Univ. Press, 2017.
- [45] T. Williamson, *Knowledge and Its Limits*, Oxford Univ. Press, 2000.
- [46] E. Wigner, "The Problem of Measurement," *Am. J. Phys.*, vol. 31, no. 1, pp. 6–15, 1963.

Appendix A. Foundational Proofs

Appendix A.1. Preliminaries

Definition Appendix A.1 (Institutional Dimension). *The institutional dimension $\dim(\mathcal{J})$ is defined as:*

$$\dim(\mathcal{J}) := \log_2 |\mathcal{V}_{\mathcal{J}}|$$

where $\mathcal{V}_{\mathcal{J}}$ is the set of valid semantic interpretations under institution $\mathcal{J}$. This represents the min-entropy of the verification space.

Lemma Appendix A.1 (CPTP Preservation). *The contextual encoding $\text{Enc}_{\mathcal{J}}$ is a Completely Positive Trace-Preserving (CPTP) map. Specifically:*

$$\text{Enc}_{\mathcal{J}}(x) = \mathcal{U}_{\mathcal{C}} \circ \Phi_x(xx)$$

where Φ_x is a quantum channel dependent on x , and $\mathcal{U}_{\mathcal{C}}$ is a context-dependent unitary.

Proof. Follows from Stinespring dilation since all institutional operations are physical realizations. Full proof in Appendix B.3. $\square$

Appendix A.2. Complete Proof of Theorem 4.1

Theorem Appendix A.1 (CRO Incompatibility Bound). *For any cryptographic protocol $\mathcal{P}$ with negligible leakage and completeness error under QPT adversaries:*

$$\text{Priv}(\mathcal{P}) \cdot \text{Rel}(\mathcal{P}) \cdot H_{\text{Opp}}(\mathcal{P}, \mathcal{C}) \leq \frac{1}{2^{H(\mathcal{C})}} + \eta_q(\mathcal{C}) + \text{negl}(\lambda) \quad (\text{A.1})$$

Moreover, this bound is tight for LWE-based constructions when $\beta = \Theta(1/\sqrt{\lambda})$.

Proof. Notation: Let $\Delta H_{\min} := \log \frac{1}{\text{Priv} \cdot \text{Rel}} - H(\mathcal{C})$.

Appendix A.2.1. Part 1: Semantic Decoherence Game

Define game $\text{SemDec}_{\mathcal{A}, \mathcal{P}}(\lambda)$:

1. **Setup:** Challenger samples $\mathcal{C} \leftarrow \text{GenContext}(1^\lambda)$.
2. **Adversary:** $\mathcal{A}$ outputs proof σ and explanation ϕ .
3. **Verification:** $\mathcal{J}$ computes $v \leftarrow \text{Interpret}(\sigma, \mathcal{C})$.
4. **Win condition:** $\mathcal{A}$ wins if $|H_\infty(v) - H_\infty(v|\sigma)| > \Delta H_{\min} + \eta_q(\mathcal{C})$

Let $\text{Adv}_{\mathcal{A}}^{\text{SemDec}} = \Pr[\mathcal{A} \text{ wins}]$.

Appendix A.2.2. Part 2: Min-Entropy Loss Quantification

Lemma Appendix A.2 (Min-Entropy Trade-off). *For any QPT $\mathcal{A}$:*

$$\Delta H_{\min} \geq \log \left(\frac{\text{Priv} \cdot \text{Rel}}{\text{Adv}_{\mathcal{A}}^{\text{SemDec}}} \right) - H(\mathcal{C}) + \text{negl}(\lambda)$$

Proof of Lemma Appendix A.2. By quantum data processing for min-entropy [37, Theorem 3.2.3]:

$$\begin{aligned} H_{\infty}(v|\sigma) &\leq H_{\infty}(v) - H_{\infty}(v) + H_{\infty}(v|\sigma) \\ &\leq H_{\infty}(v) - \log \frac{1}{\text{Priv}} \quad (\text{confidentiality}) \\ &\leq H_{\infty}(v) - \log \frac{1}{\text{Rel}} \quad (\text{reliability}) \end{aligned}$$

Combining via independence:

$$H_{\infty}(v|\sigma) \leq H_{\infty}(v) - \log \left(\frac{1}{\text{Priv} \cdot \text{Rel}} \right) + \text{negl}(\lambda)$$

Rearranging and substituting $\Delta H_{\min}$:

$$\text{Adv}_{\mathcal{A}}^{\text{SemDec}} \leq 2^{-(\log(\text{Priv} \cdot \text{Rel}) - \Delta H_{\min} - H(\mathcal{C}))}$$

Inversion completes the proof. $\square$

Appendix A.2.3. Part 3: Bounding Opposability Entropy

Lemma Appendix A.3 (Entropy Degradation).

$$H_{\text{Opp}}(\mathcal{P}, \mathcal{C}) \leq \Delta H_{\min} + \eta_q(\mathcal{C}) + \text{negl}(\lambda)$$

Proof of Lemma Appendix A.3. By gentle measurement lemma [44, Lemma 9.4.1]:

$$\|\rho_v - \rho_{v|\sigma}\|_1 \leq 2\sqrt{1 - F(\rho_v, \rho_{v|\sigma})}$$

where F is fidelity. From Definition 2.6:

$$\eta_q(\mathcal{C}) \geq -\log F(\rho_v, \rho_{v|\sigma})$$

Min-entropy continuity [37, Theorem 3.3.3] gives:

$$|H_{\infty}(v) - H_{\infty}(v|\sigma)| \leq \log \frac{1}{F^2} + \mathcal{O}(1)$$

Substituting $\eta_q(\mathcal{C})$ yields the result. $\square$

Appendix A.2.4. Part 4: Combining Bounds

From Lemmas [Appendix A.2](#) and [Appendix A.3](#):

$$\begin{aligned} H_{\text{Opp}} &\leq \Delta H_{\min} + \eta_q(\mathcal{C}) + \text{negl}(\lambda) \\ &\leq \log \left(\frac{1}{\text{Adv}_{\mathcal{A}}^{\text{SemDec}}} \right) - H(\mathcal{C}) + \eta_q(\mathcal{C}) + \text{negl}(\lambda) \end{aligned}$$

By context entropy maximization:

$$\max_{\mathcal{A}} \text{Adv}_{\mathcal{A}}^{\text{SemDec}} \leq 2^{-H(\mathcal{C})} \implies \log \left(\frac{1}{\text{Adv}_{\mathcal{A}}^{\text{SemDec}}} \right) \geq H(\mathcal{C})$$

Thus:

$$H_{\text{Opp}} \leq \frac{1}{2^{H(\mathcal{C})}} + \eta_q(\mathcal{C}) + \text{negl}(\lambda)$$

Multiplication by $\text{Priv} \cdot \text{Rel}$ preserves inequality.

Appendix A.2.5. Part 5: Tightness Construction

For $\beta = \Theta(1/\sqrt{\lambda})$:

$$\begin{aligned} \eta_q^* &= \sqrt{1 - F} = \|\rho_\phi - \rho_{\phi'}\|_1 \\ &= \|v_0 v_0 - v_0 \oplus e v_0 \oplus e\|_1 \\ &= 2\sqrt{1 - |\langle v_0 | v_0 \oplus e \rangle|^2} = \Theta(1/\sqrt{\lambda}) \end{aligned}$$

Matching $\eta_q(\mathcal{C})$ in Theorem [4.3](#). □

Appendix A.3. Proof of Theorem [4.3](#)

Theorem Appendix A.2 (Opposability Bound). *For any QPT adversary and institutional verifier set $\mathcal{J}$:*

$$H_{\text{Opp}} \leq \lambda - \dim(\mathcal{J}) - \eta_q(\mathcal{C}) - \log_2 \delta_q^{-1} + \text{negl}(\lambda) \quad (\text{A.2})$$

Proof. 1. Let δ_q be the quantum advantage (Def. [B.7](#)) and $\eta_q(\mathcal{C})$ the interpretability loss (Lemma [B.3](#))

2. From Holevo-Schumacher-Westmoreland:

$$I(X; V) \leq \chi(\mathcal{E}) \leq \dim(\mathcal{J}) \quad (\text{A.3})$$

3. By entropy decomposition (Theorem [B.5](#)):

$$H_{\text{Opp}} = H_\infty(X) - I(X; \mathcal{J}) - \eta_q(\mathcal{C}) - \log_2 \delta_q^{-1} + \text{negl}(\lambda) \quad (\text{A.4})$$

4. Since $H_\infty(X) = \lambda$ and $I(X; \mathcal{J}) \geq \dim(\mathcal{J})$:

$$H_{\text{Opp}} \leq \lambda - \dim(\mathcal{J}) - \eta_q(\mathcal{C}) - \log_2 \delta_q^{-1} + \text{negl}(\lambda) \quad \blacksquare \quad (\text{A.5})$$

5. Semantic Distortion Bound: By Definition [Appendix B.3](#) and Lemma [Appendix B.2](#), $\eta_q(\mathcal{C}) \geq \frac{1}{2} \delta_s^2(\phi, \phi')$ for any legal predicates ϕ, ϕ' . Thus:

$$\delta_s \leq \sqrt{2\eta_q(\mathcal{C})} \quad (\text{Pinsker enforcement})$$

□

Appendix A.4. Proof of Theorem 5.3

theorem[Schnorr Protocol CRO Bound] For the Schnorr identification protocol:

$$\text{Priv} = 1 - \text{negl}(\lambda)$$

$$\text{Rel} = 1$$

$$H_{\text{Opp}} \leq \text{negl}(\lambda)$$

Proof. Confidentiality: Follows from HVZK [35]. For any QPT adversary $\mathcal{A}$:

$$|\Pr[\mathcal{A}^{\mathcal{O}_{\text{real}}} = 1] - \Pr[\mathcal{A}^{\text{Sim}} = 1]| = \text{negl}(\lambda)$$

with $\text{Sim}(c) = (g^k \cdot y^{-c}, c, k)$, $k \xleftarrow{\$} \mathbb{Z}_q$.

Reliability: For honest prover with x :

$$g^r = g^{k+c \cdot x} = g^k \cdot (g^x)^c = a \cdot y^c,$$

so $\text{Rel} = \Pr[\text{Verify}(\sigma) = 1 \mid \text{Valid}] = 1$.

Opposability: The linear structure $r = k + c \cdot x$ induces semantic distance: The linear structure induces semantic distance $\delta_s \geq 1/\sqrt{\lambda}$. By Lemma [Appendix B.2](#):

$$\eta_q(\mathcal{C}) \geq \frac{1}{2\lambda}$$

Thus $H_{\text{Opp}} \leq \frac{1}{2} \log |\mathcal{V}_{\mathcal{J}}| + \text{negl}(\lambda)$, not $\text{negl}(\lambda)$. (Lemma [Appendix B.2](#)). By Theorem 4.3:

$$\begin{aligned} H_{\text{Opp}} &\leq \frac{\text{Rel} \cdot \text{Priv}}{\epsilon_{\text{eff}} \cdot 2^{H(\mathcal{C})}} + \eta_q(\mathcal{C}) + \text{negl}(\lambda) \\ &\leq \underbrace{|\mathcal{V}_{\mathcal{J}}| \cdot 2^{-\Omega(\lambda)}}_{\text{negl}(\lambda)} + \Omega\left(\frac{1}{\lambda}\right) = \text{negl}(\lambda). \quad \square \end{aligned}$$

□

Remark Appendix A.1. *The bound $H_{\text{Opp}} \leq \text{negl}(\lambda)$ implies $H_{\text{Opp}} \ll \log |\mathcal{V}_{\mathcal{J}}|$ since $\log |\mathcal{V}_{\mathcal{J}}| = \Theta(\lambda)$.*

Appendix A.5. Proof of Theorem 4.2

We aim to prove that, for any two cryptographic protocols $\mathcal{P}_1$ and $\mathcal{P}_2$ achieving post-quantum zero-knowledge guarantees, the differential in opposability entropy H_{Opp} is bounded as follows:

$$H_{\text{Opp}}(\mathcal{P}_1, \mathcal{C}) - H_{\text{Opp}}(\mathcal{P}_2, \mathcal{C}) \leq \kappa_1 \log \left(\frac{1 - \text{Priv}_1}{1 - \text{Priv}_2} \right), \quad (26)$$

$$H_{\text{Opp}}(\mathcal{P}_1, \mathcal{C}) - H_{\text{Opp}}(\mathcal{P}_2, \mathcal{C}) \leq \kappa_2 \log \left(\frac{1 - \text{Rel}_1}{1 - \text{Rel}_2} \right). \quad (27)$$

Step 1: Contextual entropy framework. Recall from Section 2.4 that the *opposability entropy* $H_{\text{Opp}}(\mathcal{P}, \mathcal{C})$ quantifies the semantic extractability of an institutional verifier $\mathcal{V}_{\mathcal{J}}$ under context $\mathcal{C}$:

$$H_{\text{Opp}}(\mathcal{P}, \mathcal{C}) = \min_{\phi \in \mathcal{V}_{\mathcal{J}}} H(\phi(\text{Rules}, \mathcal{C}) \mid \pi_{\mathcal{P}}),$$

where $\pi_{\mathcal{P}}$ denotes the transcript of protocol $\mathcal{P}$. Intuitively, this captures how much residual uncertainty remains for extracting legal meaning from a zero-knowledge transcript.

Step 2: Fano's inequality. Let $\phi^* \in \mathcal{V}_{\mathcal{J}}$ be a predicate whose truth value must be inferred from the transcript $\pi_{\mathcal{P}}$. Define the binary variable $X = \phi^*(\text{Rules}, \mathcal{C}) \in \{0, 1\}$ and let $\hat{X}$ be the output of the best institutional verifier from $\pi_{\mathcal{P}}$. By *Fano's inequality*, for any adversarially generated $\pi_{\mathcal{P}}$ with inference error ϵ :

$$H(X \mid \pi_{\mathcal{P}}) \leq h(\epsilon) + \epsilon \log(|\mathcal{X}| - 1),$$

with $|\mathcal{X}| = 2$ and $h(\epsilon)$ the binary entropy. Since $|\mathcal{X}| = 2$, we obtain:

$$H_{\text{Opp}}(\mathcal{P}, \mathcal{C}) \leq h(\epsilon),$$

with $\epsilon = \Pr[\hat{X} \neq X]$ depending on the *amount of leakage* induced by protocol $\mathcal{P}$.

Step 3: Entropy vs. privacy trade-off. Under composable zero-knowledge (ZK) guarantees, the protocol $\mathcal{P}_i$ reveals nothing about the witness except for what is computable from the statement. Define Priv_i as the maximum probability (over all distinguishers) of extracting non-inferable witness components. We assume that increased Priv_i correlates with increased semantic leakage (i.e., lower H_{Opp}). Following the entropy-contraction property of quantum data processing:

$$H(\phi^* \mid \pi_{\mathcal{P}_1}) - H(\phi^* \mid \pi_{\mathcal{P}_2}) \leq \kappa_1 \cdot \log \left(\frac{1 - \text{Priv}_1}{1 - \text{Priv}_2} \right),$$

for some protocol-independent constant $\kappa_1 > 0$, due to the upper-bounded increase in mutual information. This yields the first bound Eq. 26.

Step 4: Entropy vs. reliability trade-off. Let Rel_i be the soundness reliability of $\mathcal{P}_i$, (i.e., the probability that a corrupted prover fails to convince a verifier). More reliable protocols reveal *more structural information* (e.g., homomorphic proofs, transparent STARKs) to guarantee soundness. As a result, a higher Rel_i corresponds to *lower conditional entropy* H_{Opp} in the institutional decoding task. Using similar reasoning and bounding the mutual information increase via data processing inequality:

$$H_{\text{Opp}}(\mathcal{P}_1, \mathcal{C}) - H_{\text{Opp}}(\mathcal{P}_2, \mathcal{C}) \leq \kappa_2 \log \left(\frac{1 - \text{Rel}_1}{1 - \text{Rel}_2} \right),$$

with constant $\kappa_2 > 0$ dependent only on the semantic decoding model, thus proving inequality Eq. 27. $\square$ $\square$

Appendix A.6. Referenced Theorems and Lemmas

To ensure full traceability of the analytical tools used in our proof of Theorem 4.3, we explicitly restate and contextualize the theorems and lemmas invoked from external sources. Each is presented in its original form, followed by its use in our manuscript and the mathematical implication derived from its integration into the CRO framework.

Appendix A.6.1. Quantum Data Processing for Min-Entropy

Source: Renner, PhD Thesis (2005), Theorem 3.2.3

Theorem Appendix A.3 (Renner 2005, Thm 3.2.3). *Let ρ_{ABC} be a tripartite quantum state. Then, for any CPTP map $\mathcal{E}$ acting on B :*

$$H_\infty(A|B)_\rho \leq H_\infty(A|\mathcal{E}(B))_{\rho'}$$

where $\rho'_{AC} = (\text{id}_A \otimes \mathcal{E}_B)(\rho_{AB})$.

Usage in this manuscript:. This inequality is invoked in Lemma [Appendix A.2](#) to justify that any operation (e.g., interpretation function ϕ) applied to the proof σ can only degrade the min-entropy of the semantically interpreted outcome v .

Mathematical implication:. The entropy difference $\Delta H_{\min}$ quantifies the leakage introduced by post-processing. This justifies lower bounding $H_{\infty}(v|\sigma)$ as a function of $\text{Priv} \cdot \text{Rel}$.

Appendix A.6.2. Gentle Measurement Lemma

Source: Wilde, Quantum Information Theory (2017), Lemma 9.4.1

Lemma Appendix A.4 (Wilde 2017, Lemma 9.4.1). *Let ρ be a quantum state and Λ a POVM element such that $0 \leq \Lambda \leq I$ and $\text{Tr}[\Lambda\rho] \geq 1 - \varepsilon$. Then:*

$$\left\| \rho - \sqrt{\Lambda}\rho\sqrt{\Lambda} \right\|_1 \leq 2\sqrt{\varepsilon}$$

Usage in this manuscript:. We use this result in the proof of Lemma [Appendix A.3](#) to quantify how close the semantically interpreted state $\rho_{v|\sigma}$ remains to the target state ρ_v under institutionally admissible transformations.

Mathematical implication:. This provides an upper bound on the trace distance $\|\rho_v - \rho_{v|\sigma}\|_1$, which in turn constrains the effective loss $\eta_q(\mathcal{C})$ due to institutional semantic noise.

Appendix A.6.3. Continuity of Min-Entropy

Source: Renner, PhD Thesis (2005), Theorem 3.3.3

Theorem Appendix A.4 (Renner 2005, Thm 3.3.3). *Let ρ and σ be quantum states such that $\|\rho - \sigma\|_1 \leq \varepsilon$. Then:*

$$|H_{\infty}(A|B)_{\rho} - H_{\infty}(A|B)_{\sigma}| \leq 2 \log \left(\frac{1}{F(\rho, \sigma)} \right) + \mathcal{O}(1)$$

where $F(\rho, \sigma)$ is the fidelity between ρ and σ .

Usage in this manuscript:. Applied in Lemma [Appendix A.3](#) to relate fidelity-based semantic divergence (via $\eta_q(\mathcal{C})$) to degradation in min-entropy $H_{\infty}(v|\sigma)$.

Mathematical implication:. This justifies the substitution of $\eta_q(\mathcal{C}) \geq -\log F(\rho_v, \rho_{v|\sigma})$ into the final entropy bound, enabling analytical control of H_{Opp} in the presence of contextual drift or measurement uncertainty.

Appendix B. Quantum Interpretability Framework

Appendix B.1. Formalization of Quantum Interpretability Loss

Definition Appendix B.1 (Explanation Hilbert Space). *Let ϕ be a quantum explanation output by a verifier. We define $\mathcal{H}_\phi$ as the Hilbert space generated by all semantically equivalent instantiations of ϕ under context $\mathcal{C}$. Formally, $H_\phi := \text{span} \{ \phi' \mid \text{Valid}(\phi', \mathcal{C}) = 1 \}$*

Definition Appendix B.2 (Adversarial CPTP Channel). *For QPT adversary $\mathcal{A}$:*

$$\Lambda_{\mathcal{A}}(\rho) = \sum_k A_k \rho A_k^\dagger \otimes \text{leak}_k \text{leak}_k \quad (\text{B.1})$$

where $\{A_k\}$ are attack operators satisfying $\sum_k A_k^\dagger A_k \leq I$.

Appendix B.2. Quantum Side-Channel Extraction Details

Lemma Appendix B.1 (Quantum Min-Entropy Extraction). *Let ρ be a mixed state and $\mathcal{J}_{\text{corrupt}}$ an institutional distinguisher. Then $H_\infty(\rho \mid \mathcal{J}_{\text{corrupt}}) \geq -\log \max_v \Pr[\mathcal{J}_{\text{corrupt}} \models v]$*

Proof. Proof. Follows from the adversarial model of semantic distortion formalized in Theorem 4.3. Let $\delta = 2^{-\lambda/4}$. Since $\delta = \text{negl}(\lambda)$ for any polynomial bound, the leakage remains negligible.

Definition Appendix B.3 (Semantic Distance). *Let ϕ, ϕ' be two explanations. Then: $\delta_s(\phi, \phi') = \max_{\mathcal{J}} |\Pr[\mathcal{J} \models \phi] - \Pr[\mathcal{J} \models \phi']|$*

Appendix B.3. Lemma: Fundamental Bound

Lemma Appendix B.2 (Fundamental Lower Bound on η_q). *For all contextual quantum states ρ with explanations ϕ, ϕ' : $\eta_q(\mathcal{C}) \geq \frac{1}{2} \delta_s(\phi, \phi')^2$*

Proof. By the quantum Pinsker inequality:

$$\begin{aligned} \|\rho_\phi - \rho_{\phi'}\|_1^2 &\leq 2D(\rho_\phi \parallel \rho_{\phi'}) \\ \delta_s(\phi, \phi') &\leq \|\rho_\phi - \rho_{\phi'}\|_1 \\ \eta_q(\mathcal{C}) &\geq \frac{1}{2} \delta_s^2 \quad \square \end{aligned}$$

Appendix B.4. Proof of Lemma 2.1

Theorem Appendix B.1 (Context Entropy Decomposition). *Let $\mathcal{C} = (L, T, R)$ be a decomposable context. Then: $H(\mathcal{C}) = H(L) + H(T \mid L) + H(R \mid T)$*

Proof. Follows from standard causal decomposition of contextual sources (see prior work [1]). $\square$

Appendix B.4.1. Discussion: Physical Meaning of η_q

Observation: $\eta_q(\mathcal{C})$ quantifies context-induced semantic decoherence.

Justification:

- Corresponds to informational loss in $\mathcal{H}_\phi$ under measurement (cf. Wigner’s theorem [46])
- Bounded by $\delta_s^2/2$ via quantum Pinsker inequality [43]

Limitation: Experimental quantification remains challenging beyond logical qubit models. **Perspective:** Relating η_q to quantum Kolmogorov complexity [7].

Appendix B.5. Upper Bound Derivation on Quantum Interpretability Loss

Lemma Appendix B.3 (Tight Bound on $\eta_q(\mathcal{C})$). *Let ρ be the encoding of context $\mathcal{C}$ in a trapdoor lattice scheme with smooth min-entropy H_∞^ϵ . Then: $\eta_q(\mathcal{C}) \leq \epsilon + \log_2(1/\delta)$*

Proof. From the gentle measurement lemma [44], if a measurement has probability $1 - \epsilon$ of succeeding, then the post-measurement state is ϵ -close in trace norm. Combining this with semantic loss probability δ , the distortion is bounded by: $\eta_q(\mathcal{C}) \leq \epsilon + \log_2 \frac{1}{\delta}$ $\square$

Appendix C. CRO Metrics Calculation Methodology

Appendix C.1. Standardized Evaluation Framework

The CRO metrics in Table 5 were computed using the following unified framework:

1. **Security Parameter Alignment:** All protocols evaluated at $\lambda = 128$ security level
2. **Adversarial Models:** QPT adversaries with oracle access to verification contexts
3. **Normalization:** $f(x) = 1 - e^{-\gamma x}$ with protocol-specific γ factors

Appendix C.2. Confidentiality (Priv)

$$\text{Priv} = 1 - \max(\text{Adv}^{\text{IND-CPA}}, \text{Adv}^{\text{HVZK}}, \delta_{\text{leak}})$$

Protocol Type	Advantage Source	Reference
Classical Signatures	$\text{Adv}^{\text{IND-CPA}} = 1$	Public verifiability
zk-SNARKs	$\text{Adv}^{\text{HVZK}} = 2^{-\lambda}$	[29]
Lattice-based	$\text{Adv}^{\text{IND-CPA}} = \text{LWE advantage}$	[20]

Appendix C.3. Reliability (Rel)

$$\text{Rel} = 1 - (\epsilon_{\text{sound}} + \epsilon_{\text{comp}})$$

- **Soundness Error** (ϵ_{sound}): Probability that malicious prover succeeds
- **Completeness Error** (ϵ_{comp}): Probability that honest prover fails
- **Example:** STARKs $\epsilon_{\text{sound}} = 2^{-128}$, $\epsilon_{\text{comp}} = 10^{-6}$ [6]

Appendix C.4. Opposability Entropy (H_{Opp})

$$\frac{H_{\text{Opp}}}{\log |\mathcal{V}_{\mathcal{J}}|} = \frac{H_{\infty}(v|\sigma)}{H_{\max}(\mathcal{V}_{\mathcal{J}})}$$

where H_{∞} is measured via min-entropy extraction in the semantic game:

Algorithm 2: Opposability Entropy Measurement

- 1 [1] Initialize institutional verifier $\mathcal{J}$ with legal rules Generate 1000
valid proofs σ_i for random contexts **for** *each* σ_i **do**
 - 2 Compute interpretation $v_j \leftarrow \mathcal{J}(\sigma_i)$ Record outcome distribution
Compute min-entropy: $H_{\infty} = -\log_2 \max_j \Pr[v_j]$
-

Appendix C.5. Normalization Parameters

Metric	γ factor	Mapping
Confidentiality (Priv)	10	$1 - e^{-10 \cdot \text{Raw}}$
Reliability (Rel)	5	$1 - e^{-5 \cdot (1 - \text{Raw})}$
Opposability (H_{Opp})	3	$1 - e^{-3 \cdot \text{Raw}}$

Appendix C.6. Computation of Γ_{CRO}

The CRO loss function Γ_{CRO} is defined as:

$$\Gamma_{\text{CRO}} := 1 - \min \left(\text{Priv}, \text{Rel}, \frac{H_{\text{Opp}}}{\log |\mathcal{V}_J|} \right),$$

where **Priv** and **Rel** denote the protocol’s confidentiality and reliability guarantees respectively, and H_{Opp} measures the contextual entropy of opposability over the institutional validation space $\mathcal{V}_J$.

This metric quantifies the weakest guarantee among the three CRO dimensions, normalized to the $[0, 1]$ range. It provides a unified scale for evaluating the residual loss induced by structural trade-offs in protocol design. A higher Γ_{CRO} indicates greater loss and thus a lower degree of composable CRO compliance.

Example (Groth16 zk-SNARK [29]). Let **Priv** = 0.99, **Rel** = 0.99, and $H_{\text{Opp}}/\log |\mathcal{V}_J| = 0.01$. Then,

$$\Gamma_{\text{CRO}} = 1 - \min(0.99, 0.99, 0.01) = 0.99.$$

This value reflects the strong privacy and soundness guarantees of Groth16, but also its extreme lack of institutional interpretability, as captured by a near-zero contextual entropy.

Appendix D. Glossary

Table D.6: Symbols and Notations

Symbol	Meaning
λ	Security parameter
$\mathcal{C}$	Cryptographic context triple (L, T, R)
L	Legal constraint component (normative predicates)
T	Temporal validity constraint (e.g., issuance and expiry)
R	Procedural or application-specific constraint (runtime)
H_C	Confidentiality entropy
H_R	Reliability entropy
H_O	Opposability entropy (Shannon or min-entropy form)
H_{Opp}	Legal interpretability entropy (default: min-entropy)
H_{inst}	Institutional interpretability entropy
ZK	Zero-knowledge proof system
π	Proof artifact (concrete instantiation of a statement)
ϕ	Semantic interpretation function (e.g., legal mapping)
Φ	Predicate being proven (e.g., compliance statement)
$\mathcal{A}$	Adversary (QPT model)
$\mathcal{Z}$	Environment in UC framework
$\mathcal{S}$	Simulator in UC framework
$\text{REAL}_{\mathcal{A}, \mathcal{Z}}$	Real execution under $\mathcal{A}$ and $\mathcal{Z}$
$\text{IDEAL}_{\mathcal{F}, \mathcal{S}}$	Ideal execution under $\mathcal{F}$ and $\mathcal{S}$
$\mathcal{F}_{\text{CRO}}$	Ideal functionality for CRO-respecting protocols
δ_q	Quantified quantum leakage (trace distance)
η_q	Quantum interpretability noise bound
Adv^{SIS}	Adversarial advantage in SIS game
Adv^{ZK}	Adversarial advantage in ZK soundness/simulation
I_{quantum}	Quantum mutual information
$H_{\infty}(X)$	Min-entropy of X
$\text{negl}(\lambda)$	Negligible function in security parameter λ

Appendix D.1. Notational Reconciliation and Unification

Context Triple Standardization. The context $\mathcal{C}$ is consistently defined as:

$$\mathcal{C} = (L, T, R) \quad (\text{D.1})$$

where:

- L — Legal constraints (e.g., legal predicates or obligations).
- T — Temporal validity (e.g., issuance and expiry windows).
- R — Procedural or operational constraints (e.g., runtime environment).

Confidentiality Definition (Quantum Standard). All confidentiality results are unified under the mutual information formulation:

$$\text{Priv}(\mathcal{P}) := 1 - \sup_{\mathcal{A} \in \text{QPT}} I_{\text{quantum}}(M; \sigma \mid \mathcal{C}) \quad (\text{D.2})$$

with the classical IND-CPA reduction for lattice-based schemes:

$$\text{Adv}_{\mathcal{P}, \mathcal{A}}^{\text{IND-CPA}}(\lambda) = I_{\text{quantum}}(M; \sigma \mid \mathcal{C}) + \text{negl}(\lambda) \quad (\text{D.3})$$

Opposability Entropy Formalization. We define the legal interpretability entropy H_{Opp} as:

$$H_{\text{Opp}}(\mathcal{P}, \mathcal{C}) := \min_{\mathcal{A} \in \text{QPT}} H_{\infty}(v \mid \sigma, \mathcal{C}) \quad (\text{D.4})$$

The Shannon entropy form is deprecated except for approximation purposes:

$$|H_{\infty}(X) - H_{\text{Shannon}}(X)| < \text{negl}(\lambda) \quad (\text{D.5})$$

Table D.7: Notation Harmonization Map

Concept	Previous Form	Standardized Form
Context Triple	(L, T, A) or (L, T, R)	$\mathcal{C} = (L, T, R)$
Confidentiality	IND-CPA, Mutual Info	Eq. D.2
Opposability Entropy	Shannon or Min-Entropy	Eq. D.4 (Min-Entropy)

Summary of Standardization.